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# Statistical Models for Earthquake Occurrences and Residual Analysis for Point Processes

YOSIHIKO OGATA*

This article discusses several classes of stochastic models for the origin times and magnitudes of earthquakes. The models are compared for a Japanese data set for the years 1885–1980 using likelihood methods. For the best model, a change of time scale is made to investigate the deviation of the data from the model. Conventional graphical methods associated with stationary Poisson processes can be used with the transformed time scale. For point processes, effective use of such *residual analysis* makes it possible to find features of the data set that are not captured in the model. Based on such analyses, the utility of seismic quiescence for the prediction of a major earthquake is investigated.

KEY WORDS: Akaike information criterion; Epidemic-type models; Conditional intensity; Likelihood; Marked point process; Seismic quiescence; Trigger models.

## 1. INTRODUCTION

It is widely accepted that some time after the occurrence of a major earthquake the aftershock activity dies off and background seismicity surpasses the aftershock activity. Prior to the next major earthquake, preseismic quiescence and then foreshocks are expected to appear in the focal region (Mogi 1968). It is not easy, however, to identify foreshocks before the occurrence of a main shock. In particular, foreshocks tend to be less numerous than aftershocks. Thus the seismic quiescence and related seismic gap have been studied by many seismologists for the purpose of earthquake prediction [e.g., see Kanamori (1981) and some other authors in the same volume].

Some have questioned the usefulness of seismic quiescence and suggested that it can be considered a mere result of the decaying activity of aftershocks from the last major earthquake (see Lomnitz 1982; Lomnitz and Nava 1983). The latter paper discusses the similarity between certain observed earthquake sequences and sequences featuring seismic quiescences generated from stochastic models that are based on simple statistical assumptions such as the Gutenberg–Richter law of magnitude frequency, Omori's decaying frequency law of aftershocks, and Båth's law.

To investigate this issue we have to investigate seismic quiescence by means of a quantitative comparison with background seismicity and aftershock activity. There have been some attempts to develop statistical techniques for the definition and detection of quiescence. For example, Habermann and Wyss (1984) developed a statistical test procedure to detect quiescences for seismic activity with data from which the effect of aftershocks is removed. Aftershocks, however, constitute the greatest proportion of shocks in an earthquake catalog, and there have been many detailed studies on aftershock sequences (e.g., see Utsu 1969, 1970, 1971, 1972). If these are effectively considered in the analysis, aftershocks can give us useful information for understanding the whole cycle of seismic activity (see Matsu'ura 1986; Ogata and Shimazaki 1984; Okada 1978).

This article selects a statistical model for the standard activity of earthquake series by comparing several possible models using likelihood methods. In the next section some possible stochastic models for seismic activity derived from empirical studies, mainly of aftershock statistics, are reviewed and discussed. In Section 3.1 an earthquake data set for which the models seem applicable is considered. Model comparisons, using this data set, are presented in Section 3.2.

Another feature of this article is the systematic use of a new kind of *residual* analysis for point process data, described in Section 3.3, which involves a change of time scale that depends on the conditional intensity rate of the estimated model. The model selected fits the data well enough that the residuals reveal certain events more clearly than the original data. This makes possible a precise quantitative definition of seismic quiescence in Section 4 as well as the detection of a stretch of outliers. Then it is demonstrated that seismic quiescence has some utility for statistical prediction of a major earthquake. The final section contains some additional discussion and some conclusions.

## 2. STATISTICAL MODELS FOR THE EARTHQUAKE PROCESS

### 2.1 Empirical Studies on Aftershock Statistics

The statistical properties of the occurrence of aftershocks have long been some of the main objects of seismological studies in connection with the processes of

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earthquake generation. Distributions of aftershocks in time, space, and magnitude are usually included in a general survey of destructive earthquakes. In this section some studies of aftershock statistics are reviewed briefly. Based on this review, some plausible point process models are subsequently developed to describe the standard activity of earthquake series.

The frequency of aftershocks per unit time interval (one day, one month, etc.) is well described by the modified Omori formula (Utsu 1961),

$$n(t) = K/(t + c)^p \qquad (K, c, p; \text{parameters}), \quad (1)$$

where $t$ is the time since occurrence of the shock. $K$ depends on the lower bound of the magnitude of aftershocks counted in $n(t)$, whereas $p$ and $c$ are known to be independent of this choice of lower bound. Utsu (1969, 1970) extensively investigated the aftershock sequences of this century's earthquakes in Japan. Plotting observed $n(t)$ versus time $t$ on a log–log-scaled plane, he showed that the graph tends to a straight line whose slope is an estimate of $p$. With such plots, for example, Utsu (1969) discovered the remarkable fact that the aftershock activity of the Nobi earthquake of 1891 continued for at least 80 years with a continuously decreasing rate of occurrence.

It is known empirically that the magnitudes of the earthquakes follow roughly an exponential distribution.

$$\Pr\{\text{Mag} > M\} = e^{-\beta M}. \quad (2)$$

In the seismological literature this relationship, called Gutenberg–Richter's law of magnitude of frequency, is most often written in the form

$$\log_{10} F(M) = a - bM, \quad (3)$$

where $F(M)$ refers to the frequency of earthquakes with magnitudes not smaller than $M$ [see Gutenberg and Richter (1944); see also Fig. 3 later in this article]. The constant $b$, called $b$ value, is then related to $\beta$ by the equation $\beta = b \cdot \log_e 10$ and commonly takes a value near unity.

Utsu (1971, pp. 420–427) studied theoretically the relation between the size of aftershock activity and the magnitude of the main shock, based on the aforementioned law of magnitude frequency (3) and some theoretical properties of order statistics. For a sequence of aftershocks triggered by a main shock with magnitude $M$, he found the relation

$$\log_{10} A = \alpha M + \text{constant}, \quad (4)$$

where $A$ is the total number of aftershocks in the sequence and $\alpha$ is a constant. From this and the magnitude frequency law (2) for the main shocks, the frequency distribution of $A$ takes the form

$$f(A) = \text{constant } A^{-q} \quad (5)$$

for some positive $q$. Empirical evidence in support of (5) was provided by plotting the logarithm of the cumulative frequency distribution of the number of aftershocks versus the magnitude of the corresponding main shock (Utsu 1971, fig. 133). For seismological readers particularly, I point out that (4) is similar in form to some widely used empirical formula like that given by Utsu and Seki (1955),

$$\log_{10} S = 1.02M - 4.0, \quad (6)$$

relating the aftershock area $S(km^2)$ and the magnitude $M$ of the main shock for shallow earthquakes in Japan and its vicinity.

Combining (1) and (4), Utsu (1970, p. 229) gave the standard equation for the rate of occurrence of aftershocks having magnitude $M$, and larger, $t$ days after a large shallow earthquake of magnitude $M_0$:

$$n(t) = 10^{.85(M_0 - M_r) - 1.83}/(t + .3)^{1.3} \quad (7)$$

per day.

For constructing statistical models of earthquake occurrence, some authors (Lomnitz 1966; Lomnitz and Nava 1983; Utsu 1972) assumed that there was no detailed causality or interaction between the occurrence times of aftershocks, so the time sequence of aftershocks could be treated as a nonstationary Poisson process. Vere-Jones (1975) suggested that this assumption is mainly due to the analysis of Jeffreys (1938), in which daily numbers of aftershocks of the Tango earthquake of 1927 were tested by evaluating chi-squared statistics to establish the independence of the occurrence numbers. A similar result was obtained in Lomnitz and Hax (1966) by analyzing autocorrelations. I will argue, however, that these analyses do not really establish the independence of the precise times at which the aftershocks occur. Furthermore, the so-called secondary aftershocks, or the aftershocks of an aftershock (see Utsu 1970, pp. 214–216), are not explained by such models. According to Ogata (1983) the aftershocks of the Tango earthquake had at least two secondary sequences. In addition, a few analyses suggesting the interdependence of aftershocks can be obtained by using the theory of runs tests (Utsu 1970, pp. 217–229).

Although the independence of the occurrence of main shocks has been assumed in many models, some papers discuss the migration of large earthquakes and the causal relationships between seismic activity in different geophysical regions (Mogi 1968, 1973; Ogata and Katsura 1986; Utsu 1975). Many such reports have been given constantly at the meetings of the Seismological Society of Japan.

## 2.2 Trigger Models

The model discussed by Lomnitz and Nava (1983) is related to the trigger model suggested by Vere-Jones and Davies (1966), which Vere-Jones (1970) described as a special case of the Neyman–Scott clustering model. This model assumes a series of primary events (main shocks) distributed completely randomly in time. Each of these primary events may generate a secondary series of events (aftershocks). It is assumed that the conditional probability that an aftershock (of magnitude above a certain level) will occur in the small time interval $(t, t + dt)$, triggered by a main shock at time $t_0$, is equal to $\sigma_{t_0}(t)\, dt$

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with

$$\sigma_{t_0}(t) = \xi \cdot f(t - t_0), \qquad t \geq t_0$$
$$= 0, \qquad t < t_0, \tag{8}$$

where $f(t)$ is a normalized function; that is, $\int_0^\infty f(t)\,dt = 1$ and $\xi$ is the average number of secondary events produced by a primary event at $t_0$.

Vere-Jones and Davies (1966) suggested that the inverse power type of decay function

$$f(t) = (p - 1)c^{p-1}/(c + t)^p, \tag{9}$$

which arises from normalizing (1), gives a better fit than the negative exponential function. They assumed further that $\xi$ itself is a random variable with finite mean and variance, probably as a technical requirement for the estimation of the parameters using the second-moment properties. In Lomnitz and Nava (1983), $\xi$ is taken to be proportional to $M_m - M_r$, where $M_m$ is the magnitude of the main shock and $M_r$ is the cutoff magnitude, which together with the law of magnitude frequency implies that $\xi$ has a negative exponential distribution. In this article, I will consider only a restricted form of trigger model, which will be described later.

### 2.3 Epidemic-Type Model

Another type of model appeared in applications to population genetics. Kendall (1949) introduced an age-dependent birth and death process such that for any individual of age $x$ alive at time $t$, for the next interval $(t, t + dt)$ there are probabilities $g(x)\,dt$ of a birth and $h(x)\,dt$ of a death, independently for each individual. Hawkes (1971) considered the self-exciting process, which is a birth process [i.e., $h(x) = 0$] allowing immigration at a rate $\mu$ per unit time. He defined the process by means of the conditional intensity rate

$$\begin{aligned}\lambda(t) &= \mathrm{E}[dN(t) \mid \text{history of } N(s) \text{ at time } t]/dt \\ &= \lim_{\Delta \to 0} \Delta^{-1} \\ &\quad \times \Pr\{\text{event in } (t, t + \Delta) \mid \text{history of} \\ &\qquad N(s) \text{ at time } t\} \\ &= \mu + \sum_{t_i < t} g(t - t_i) = \mu + \int_0^t g(t - s)\,dN(s),\end{aligned} \tag{10}$$

where $N(t)$ is the cumulative number of events, $\{t_i\}$ in $(0, t]$. This process may also be viewed as a cluster process, different from the Neymann–Scott type, in which the process $N_c(t)$ of birth times of cluster centers [see Eq. (18)] is a Poisson process of rate $\mu$ formed by the arrival of immigrants. Associated with each event of $N_c(t)$ we have a cluster of subsidiary events formed by the births of all of the descendants of all generations of the immigrant (Hawkes and Oakes 1974). One of the differences between this model and the trigger model is that the latter includes only the first generation offspring whereas in the former all events have the possibility of possessing offspring.

Extending (10) to a multivariate point process, $\{t_i^m\}$, Hawkes (1971) also defined the mutually exciting model

$$\begin{aligned}\lambda_j(t) &= \mathrm{E}[dN_j(t) \mid \text{histories of } N_m(s) \\ &\qquad \text{for all } m \text{ at time } t]/dt \\ &= \mu_j + \sum_m \sum_{t_i^m < t} g_{jm}(t - t_i^m) \\ &= \mu_j + \sum_m \int_0^t g_{jm}(t - s)\,dN_m(s)\end{aligned} \tag{11}$$

for the discrete magnitude values of $j$ and $m$. If we assume that the simplest forms $g_{jm}(t) = c(m)g_j(t)$ and consider the superposition $N(t) = \sum_m N_m(t)$ of the point process components, then the conditional intensity $\lambda(t) = \sum_j \lambda_j(t)$ is given by

$$\lambda(t) = \mu + \sum_{t_i < t} c(m_i)g(t - t_i), \tag{12}$$

where $t_i$ is the occurrence time of the superposition $N(t)$, $m_i$ is the corresponding magnitude of $t_i$, and $g(t) = \sum_j g_j(t)$. Further, $\mu = \sum \mu_m$ can be considered as a base rate that prevents the process from dying out. The model (12) coincides with the tagged Klondike-type model described in Lomnitz (1974) for earthquake series $\{(t_i, m_i)\}$ with $m_i \geq M_r$, where $M_r$ is the cutoff magnitude. Lomnitz suggested the use of

$$g(t) = ae^{-\alpha t} \tag{13}$$

in view of Boltzman's theory of elastic aftereffect. Here I would also like to consider the model

$$g(t) = K/(t + c)^p, \tag{14}$$

which corresponds to (1). A technical extension of (13),

$$g(t) = \sum_{k=1}^{\kappa} a_k t^{k-1} e^{-\alpha t}, \tag{15}$$

is proposed in Ogata and Akaike (1982) and also in Vere-Jones and Ozaki (1982). The pioneering application of the model (8) (including a trigger model as a special case) to earthquake data was carried out by Hawkes and Adamopoulos (1973), where a mixture of two exponentials is considered as an extension of (10) and a certain approximated log-likelihood is used to fit the model.

For the $c(m)$ in (12) I propose the use of

$$c(m) = e^{\beta(m - M_r)} \tag{16}$$

rather than $\beta(m - M_r)$ as given for $\xi$ in (8) by Lomnitz and Nava (1983), since (16) seems to be consistent with (4)–(7) for the descendants of the first generation (primary aftershocks). The parameter $\beta$ here measures the effect of magnitude in the production of descendants and is useful in characterizing the earthquake sequences quantitatively in relation to the classification into seismic types made by Mogi (1963) and Utsu (1970) (see also Ogata 1987). For example, earthquake swarms have small $\beta$ values, and a

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value of $\beta$ around 2 might be used as a standard for the aftershock activity in Japan and its vicinity [see Eqs. (6) and (7)].

### 2.4 Parameterization of the Models

To calculate log-likelihoods we need to describe the models in terms of the parameterized conditional intensity. Here a class of competitive models is summarized in such terms. First a unified form for the epidemic type of models is provided, as follows:

$$\lambda(t) = \mu + \sum_{t_i<t} g(t - t_i)e^{\beta(m_i - M_r)} = \mu + \int_{M_r}^{\infty}\int_0^t g(t - s)e^{\beta(m-M_r)}\, dN(s, m), \quad (17)$$

where $\lambda(t)$ is the conditional intensity for the occurrence times of the point process, $M_r$ is the cut off magnitude, and $dN(s, m)$ gives mass 1 to each shock $(t_i, m_i)$.

Thus the parametric models (17) will be compared, with $g(t - s)$ defined by (14) or by (15) for integer values of $\kappa$ [(13) is a special case in an interval]. In addition, we can examine the case in which $\beta = 0$ to determine whether the magnitude really affects the cluster size of the corresponding aftershocks; see also the last paragraph in Section 2.3.

Now I motivate the "restricted" trigger models, which will be considered subsequently. The estimation and comparison among the original trigger models was carried out by the methods using only second-moment properties, such as goodness of fit of some graphical statistics like the variance–time curve, the hazard function, the periodogram, and so on (see, e.g., Vere-Jones 1970; Vere-Jones and Davies 1966). The actual likelihood of such Neyman–Scott types of cluster models can only be calculated by an extraordinarily complicated computation allowing all possible choices of $t_i$ for cluster centers (see, e.g., Baudin 1981) unless each shock $(t_i, m_i)$ is identified in advance as a main shock or aftershock (foreshocks here are included in main shocks). The conventional approach to such identification is usually based on the observation of spatiotemporal plots. When the application of the trigger model is applied to such an identified data set, we call the model a *restricted trigger model.* That is to say, shocks are given by the decomposition

$$dN(t, m) = dN_c(t, m) + dN_a(t, m), \quad (18)$$

where $N_c$ and $N_a$ stand for the primary and secondary events, respectively. Assuming that the distribution of main shocks is stationary and random with rate $\mu$, we have

$$\lambda_c(t) = \mu. \quad (19)$$

Corresponding to (8), the conditional intensity rate of an aftershock is given by

$$\lambda_a(t) = \int_{M_r}^{\infty}\int_0^t g(t - s)e^{\beta(m-M_r)}\, dN_c(s, m) = \sum_{t_i^c<t} g(t - t_i^c)e^{\beta(m_i^c - M_r)}, \quad (20)$$

where $M_r$ is the reference magnitude. Thus the intensity of the restricted trigger model is given by the combination of (19) and (20),

$$\lambda(t) = \lambda_c(t) + \lambda_a(t) = \mu + \int_{M_r}^{\infty}\int_0^t g(t - s)e^{\beta(m-M_r)}\, dN_c(s, m) = \mu + \sum_{t_i^c<t} g(t - t_i^c)e^{\beta(m_i^c - M_r)}. \quad (21)$$

Technically, (21) can be obtained from (17) by replacing the quantities $e^{\beta(m_i^a - M_r)}$ for aftershocks $(t_i^a, M_i^a)$ by 0. I will compare (13) and (14) by using this restricted model and the epidemic model.

## 3. ANALYSIS

### 3.1 The Data and Their Features

Utsu (1982) compiled and published a complete catalog of earthquakes of magnitude 6 or more that occurred in Japan and its vicinity from 1885 through 1980. This catalog includes occurrence times, coordinates of epicenters, depths, and some other information. In addition, each shock has been classified as a main shock, a foreshock, or an aftershock. This classification was not published in Utsu (1982), but the author has kindly provided me with it privately. Thus it is also possible for us to get the log-likelihood values of the restricted trigger models (21), assuming that foreshocks are included in the category of main shocks.

I will consider shallow earthquakes of less than 100 kilometers depth with $M \geq 6.0$ in the polygonal region of the Off Tohoku district, with vertices at the points (42°N, 142°E), (39°N, 142°E), (38°N, 141°E), (35°N, 140.5°E), (35°N, 144°E), (39°N, 146°E), and (42°N, 146°E) (see Fig. 1), which is approximately between the lines of the Japan Trench and the aseismic front proposed by Yoshii (1975). This is a part of the northwestern Pacific seismic belt, where the Pacific plate is subducting beneath northeastern Japan in the Eurasian Plate, and is one of the most active seismic areas in Japan.

In this area, 483 shocks with magnitude 6 or more have occurred in the past 96 years, which means that about 5 such shocks occur on average each year. These data are listed in Table 1. Figure 2 shows the plots of the occurrence times of shocks with their magnitudes. The graphs of the cumulative number of shocks and cumulative square root of released energy are given in the same figure, where for the energy released by a shock with magnitude $M$ I have used the relation $E = 10^{11.8+1.5M}$ (erg) according to Gutenberg and Richter (1954). From Figure 2 we may assume that the seismic activity here is stationary. An actual test of this assumption is not easy, especially when the process has long-range correlation (see Fig. 7). Nevertheless, stationary models are considered here in the prior belief that such geophysical activity for a long time span should be stationary. The magnitude distribution for this region is shown in Figure 3, which clearly supports the Gutenberg–

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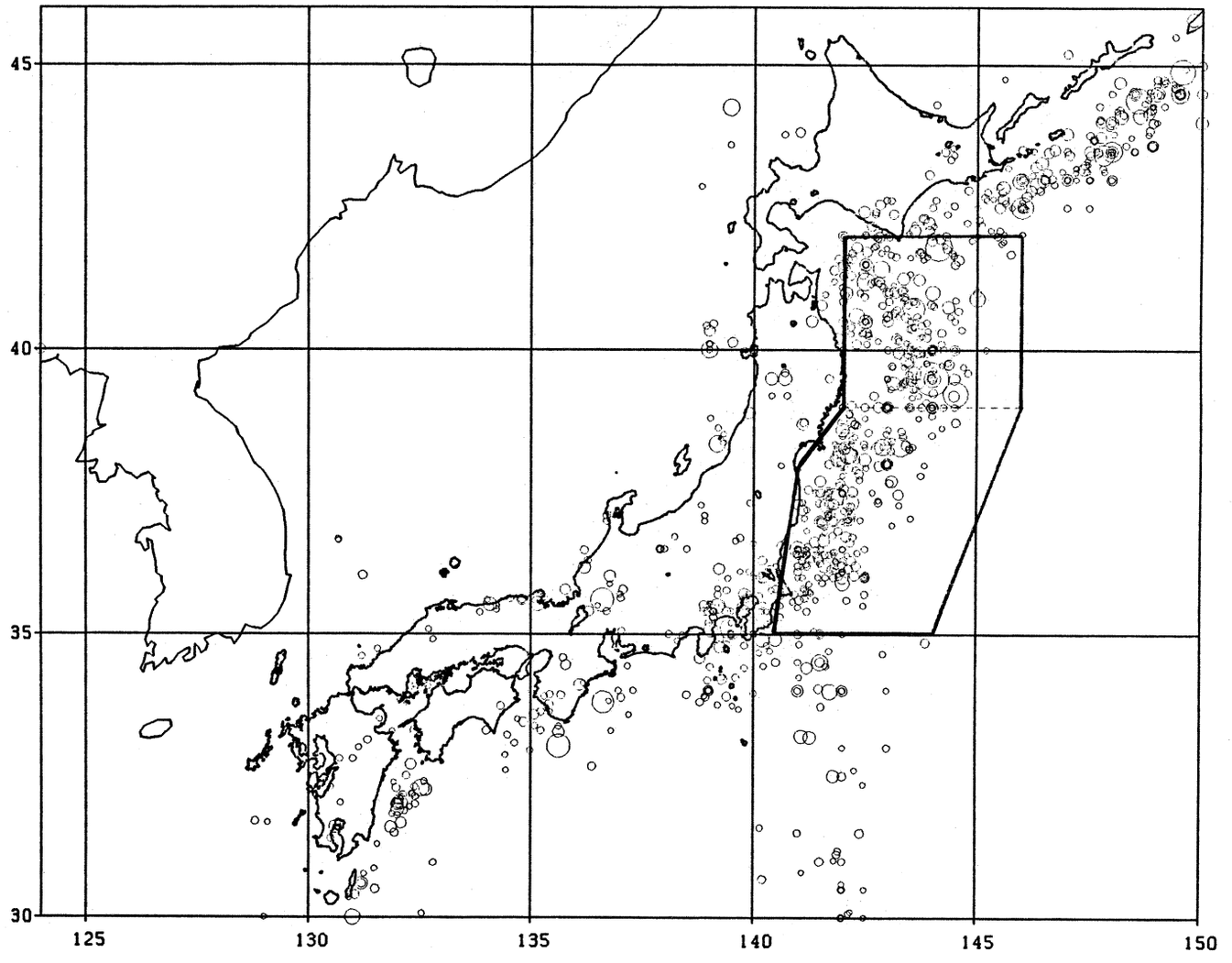



*Figure 1. The Spatial Distribution of Large Earthquakes ($M \geq 6.0$) 1895–1980 in Japan and Its Vicinity. The sizes of the open circles correspond to the sizes of the earthquakes. The data set for the polygonal region is considered.*

Richter law of magnitude frequency [see (2) and (3), Sec. 2.1]. The fact that the graph of points associated with magnitudes near 6.0 lies on the straight line suggested by the Gutenberg–Richter law demonstrates to seismologists that the catalog includes essentially all of the events of this magnitude that occurred.

Using the intervals $X_1, X_2, \ldots$ between successive earthquakes, the log-survivor function, $\log P\{X > x\}$, is plotted in Figure 4. The estimated variance–time curve, $V(s) = \text{var}\{N(t, t+s)\}$, is given in Figure 5. The empirical plot of the intensity of the so-called Palm distribution,

$$m_f(s) = \Pr\{\text{event in } (t+s, t+s+\Delta s) \mid \text{event at } t\}/\Delta s,$$

for some small $\Delta s$, is given in Figure 6. All of these plots suggest the strongly clustering nature of the occurrence times of shocks. In particular, Figure 6, with confidence error bounds assuming the Poisson model, suggests so-called long-range correlation. Indeed, if we plot an estimate of $| m_f(s) - \lambda_0 |$ ($\lambda_0$ is the mean intensity rate) on a log–log scale, we get Figure 7, which suggests inverse power decay of autocovariance at large time lags. This may support (14) rather than (13) and (15) for the time decay of the response function.

## 3.2 Comparison of Restricted Trigger Models and Epidemic Models

To compare the epidemic models with the restricted trigger models, I make use of likelihood analysis. Before writing the likelihood of the models, let us consider the assertion of Lomnitz (1982) and Lomnitz and Nava (1983) that the distribution of magnitude frequency is not affected by the occurrence times of shocks, especially just before large shocks. This means that the time series of magnitudes $\{m_i\}$ of earthquakes is independent of their occurrence times $\{t_i\}$. Based on this, the full likelihood of $\{(t_i, m_i)\}$ can be written in the following form:

$$\prod_{j=1}^{N} f_j(m_j \mid m^{(j-1)}; \eta) \prod_{j=1}^{N} g_j(t_j \mid m^{(j-1)}, t^{(j-1)}; \theta), \quad (22)$$

where $t^{(j)} = (t_1, \ldots, t_j)$ and $m^{(j)} = (m_1, \ldots, m_j)$ and

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*Table 1. Shallow Shocks ($M \geq 6.0$) in OFF Tohoku Area for 1885–1980*

| NO | YEAR | MO | DY | HR | MN | MAG | C | NO | YEAR | MO | DY | HR | MN | MAG | C | NO | YEAR | MO | DY | HR | MN | MAG | C |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1885 | 2 | 9 | 2 | 0 | 6.0 | 0 | 84 | 1908 | 1 | 15 | 21 | 56 | 6.9 | 0 | 167 | 1923 | 5 | 31 | 14 | 55 | 6.2 | 1 |
| 2 | 1885 | 6 | 11 | 9 | 20 | 6.9 | 0 | 85 | 1908 | 1 | 18 | 1 | 5 | 6.0 | 0 | 168 | 1923 | 6 | 2 | 2 | 24 | 7.3 | 0 |
| 3 | 1885 | 7 | 29 | 5 | 30 | 6.0 | 0 | 86 | 1908 | 2 | 5 | 21 | 7 | 6.0 | 0 | 169 | 1923 | 6 | 2 | 5 | 14 | 7.1 | 2 |
| 4 | 1885 | 10 | 30 | 20 | 30 | 6.2 | 0 | 87 | 1908 | 6 | 27 | 23 | 21 | 6.1 | 0 | 170 | 1923 | 6 | 7 | 2 | 36 | 6.2 | 2 |
| 5 | 1885 | 12 | 7 | 13 | 2 | 6.3 | 0 | 88 | 1908 | 11 | 22 | 16 | 15 | 6.4 | 0 | 171 | 1923 | 9 | 2 | 18 | 49 | 6.3 | 2 |
| 6 | 1885 | 12 | 19 | 18 | 26 | 6.0 | 2 | 89 | 1909 | 9 | 17 | 4 | 39 | 6.8 | 0 | 172 | 1923 | 11 | 18 | 5 | 40 | 6.3 | 1 |
| 7 | 1886 | 4 | 13 | 5 | 44 | 6.3 | 0 | 90 | 1910 | 1 | 22 | 8 | 25 | 6.0 | 0 | 173 | 1923 | 12 | 27 | 23 | 39 | 6.4 | 0 |
| 8 | 1886 | 7 | 2 | 12 | 33 | 6.3 | 2 | 91 | 1910 | 5 | 9 | 18 | 53 | 6.0 | 1 | 174 | 1924 | 2 | 3 | 7 | 25 | 6.3 | 0 |
| 9 | 1887 | 5 | 29 | 0 | 50 | 6.4 | 0 | 92 | 1910 | 5 | 10 | 22 | 56 | 6.1 | 0 | 175 | 1924 | 5 | 31 | 21 | 2 | 6.3 | 1 |
| 10 | 1887 | 5 | 29 | 1 | 10 | 6.2 | 2 | 93 | 1910 | 5 | 12 | 12 | 22 | 6.0 | 2 | 176 | 1924 | 5 | 31 | 21 | 4 | 6.4 | 1 |
| 11 | 1888 | 2 | 5 | 0 | 50 | 7.1 | 0 | 94 | 1910 | 10 | 13 | 23 | 56 | 6.3 | 0 | 177 | 1924 | 8 | 6 | 23 | 22 | 6.3 | 0 |
| 12 | 1888 | 11 | 24 | 2 | 3 | 6.5 | 0 | 95 | 1912 | 1 | 4 | 4 | 4 | 6.1 | 0 | 178 | 1924 | 8 | 15 | 3 | 2 | 7.1 | 0 |
| 13 | 1889 | 3 | 31 | 6 | 42 | 6.6 | 0 | 96 | 1912 | 1 | 9 | 6 | 21 | 6.1 | 0 | 179 | 1924 | 8 | 15 | 8 | 27 | 6.7 | 2 |
| 14 | 1890 | 11 | 17 | 9 | 31 | 6.3 | 0 | 97 | 1912 | 6 | 8 | 13 | 41 | 6.6 | 0 | 180 | 1924 | 8 | 17 | 10 | 45 | 6.3 | 2 |
| 15 | 1891 | 4 | 7 | 9 | 49 | 6.7 | 0 | 98 | 1912 | 12 | 9 | 8 | 50 | 6.6 | 0 | 181 | 1924 | 8 | 17 | 11 | 10 | 6.6 | 2 |
| 16 | 1891 | 5 | 5 | 8 | 16 | 6.2 | 0 | 99 | 1913 | 2 | 20 | 17 | 58 | 6.9 | 0 | 182 | 1924 | 8 | 25 | 23 | 31 | 6.7 | 2 |
| 17 | 1891 | 7 | 21 | 20 | 19 | 7.0 | 0 | 100 | 1913 | 5 | 22 | 5 | 36 | 6.1 | 1 | 183 | 1925 | 2 | 7 | 2 | 11 | 6.0 | 0 |
| 18 | 1892 | 10 | 22 | 19 | 9 | 6.0 | 0 | 101 | 1913 | 5 | 29 | 19 | 14 | 6.4 | 0 | 184 | 1925 | 4 | 20 | 5 | 24 | 6.3 | 0 |
| 19 | 1894 | 2 | 25 | 4 | 18 | 6.8 | 0 | 102 | 1913 | 10 | 3 | 9 | 17 | 6.1 | 1 | 185 | 1925 | 6 | 2 | 14 | 18 | 6.4 | 0 |
| 20 | 1894 | 3 | 14 | 18 | 15 | 6.0 | 2 | 103 | 1913 | 10 | 11 | 18 | 10 | 6.9 | 0 | 186 | 1925 | 11 | 10 | 23 | 44 | 6.0 | 0 |
| 21 | 1894 | 8 | 29 | 19 | 55 | 6.6 | 0 | 104 | 1913 | 10 | 13 | 2 | 5 | 6.6 | 2 | 187 | 1926 | 4 | 7 | 4 | 33 | 6.3 | 0 |
| 22 | 1894 | 11 | 28 | 1 | 5 | 7.1 | 0 | 105 | 1914 | 2 | 7 | 15 | 50 | 6.8 | 0 | 188 | 1926 | 5 | 27 | 4 | 45 | 6.4 | 0 |
| 23 | 1894 | 12 | 1 | 18 | 37 | 6.3 | 0 | 106 | 1914 | 12 | 26 | 3 | 18 | 6.1 | 0 | 189 | 1926 | 9 | 5 | 0 | 37 | 6.8 | 0 |
| 24 | 1896 | 1 | 9 | 22 | 17 | 7.5 | 0 | 107 | 1915 | 3 | 9 | 0 | 29 | 6.8 | 0 | 190 | 1926 | 10 | 3 | 17 | 25 | 6.4 | 0 |
| 25 | 1896 | 1 | 10 | 5 | 52 | 6.0 | 2 | 108 | 1915 | 4 | 6 | 5 | 25 | 6.0 | 1 | 191 | 1926 | 10 | 19 | 9 | 29 | 6.2 | 0 |
| 26 | 1896 | 1 | 10 | 11 | 25 | 6.3 | 0 | 109 | 1915 | 4 | 6 | 14 | 32 | 6.2 | 0 | 192 | 1926 | 11 | 11 | 12 | 1 | 6.1 | 2 |
| 27 | 1896 | 2 | 23 | 19 | 42 | 6.1 | 2 | 110 | 1915 | 4 | 25 | 2 | 9 | 6.4 | 0 | 193 | 1927 | 1 | 18 | 6 | 58 | 6.4 | 0 |
| 28 | 1896 | 3 | 6 | 23 | 52 | 6.0 | 2 | 111 | 1915 | 5 | 28 | 2 | 26 | 6.0 | 2 | 194 | 1927 | 3 | 16 | 15 | 52 | 6.4 | 2 |
| 29 | 1896 | 4 | 11 | 23 | 0 | 6.0 | 2 | 112 | 1915 | 6 | 5 | 6 | 59 | 6.7 | 0 | 195 | 1927 | 7 | 30 | 23 | 18 | 6.4 | 0 |
| 30 | 1896 | 6 | 15 | 19 | 32 | 8.5 | 0 | 113 | 1915 | 7 | 9 | 7 | 21 | 6.4 | 0 | 196 | 1927 | 8 | 6 | 6 | 12 | 6.7 | 0 |
| 31 | 1896 | 6 | 16 | 4 | 16 | 7.5 | 2 | 114 | 1915 | 10 | 13 | 6 | 30 | 6.8 | 0 | 197 | 1927 | 9 | 30 | 16 | 38 | 6.3 | 0 |
| 32 | 1896 | 6 | 16 | 8 | 1 | 7.5 | 2 | 115 | 1915 | 10 | 14 | 4 | 43 | 6.2 | 2 | 198 | 1928 | 5 | 27 | 18 | 50 | 7.0 | 0 |
| 33 | 1896 | 7 | 29 | 17 | 44 | 6.1 | 2 | 116 | 1915 | 10 | 15 | 1 | 28 | 6.1 | 2 | 199 | 1928 | 5 | 29 | 0 | 35 | 6.7 | 2 |
| 34 | 1896 | 8 | 1 | 11 | 49 | 6.5 | 0 | 117 | 1915 | 10 | 15 | 3 | 40 | 6.3 | 2 | 200 | 1928 | 6 | 1 | 22 | 12 | 6.5 | 2 |
| 35 | 1896 | 9 | 5 | 23 | 7 | 6.5 | 2 | 118 | 1915 | 10 | 16 | 1 | 55 | 6.0 | 2 | 201 | 1928 | 6 | 2 | 7 | 6 | 6.0 | 2 |
| 36 | 1897 | 2 | 20 | 5 | 50 | 7.4 | 0 | 119 | 1915 | 10 | 17 | 0 | 21 | 6.1 | 2 | 202 | 1928 | 8 | 1 | 4 | 28 | 6.1 | 2 |
| 37 | 1897 | 2 | 20 | 8 | 47 | 7.0 | 2 | 120 | 1915 | 11 | 1 | 16 | 24 | 7.5 | 0 | 203 | 1929 | 3 | 15 | 10 | 57 | 6.0 | 2 |
| 38 | 1897 | 3 | 27 | 19 | 49 | 6.3 | 2 | 121 | 1915 | 11 | 1 | 16 | 50 | 6.7 | 2 | 204 | 1929 | 4 | 1 | 5 | 17 | 6.3 | 0 |
| 39 | 1897 | 5 | 23 | 21 | 22 | 6.9 | 2 | 122 | 1915 | 11 | 1 | 18 | 1 | 7.0 | 2 | 205 | 1929 | 4 | 16 | 9 | 53 | 6.3 | 0 |
| 40 | 1897 | 7 | 22 | 18 | 31 | 6.8 | 0 | 123 | 1915 | 11 | 2 | 0 | 43 | 6.2 | 2 | 206 | 1929 | 5 | 31 | 9 | 10 | 6.1 | 0 |
| 41 | 1897 | 7 | 29 | 22 | 45 | 6.0 | 2 | 124 | 1915 | 11 | 4 | 12 | 13 | 6.4 | 2 | 207 | 1929 | 6 | 27 | 1 | 49 | 6.1 | 0 |
| 42 | 1897 | 8 | 5 | 9 | 10 | 7.7 | 0 | 125 | 1915 | 11 | 18 | 13 | 4 | 7.0 | 2 | 208 | 1929 | 8 | 29 | 3 | 51 | 6.3 | 0 |
| 43 | 1897 | 8 | 6 | 8 | 48 | 6.3 | 2 | 126 | 1915 | 12 | 7 | 5 | 58 | 6.5 | 0 | 209 | 1930 | 5 | 1 | 9 | 58 | 6.6 | 0 |
| 44 | 1897 | 8 | 12 | 10 | 50 | 6.1 | 2 | 127 | 1916 | 3 | 18 | 9 | 58 | 6.6 | 0 | 210 | 1930 | 8 | 21 | 19 | 44 | 6.1 | 0 |
| 45 | 1897 | 8 | 16 | 16 | 50 | 7.2 | 2 | 128 | 1916 | 5 | 15 | 8 | 56 | 6.0 | 0 | 211 | 1931 | 3 | 9 | 12 | 48 | 7.6 | 0 |
| 46 | 1897 | 10 | 2 | 21 | 45 | 6.6 | 2 | 129 | 1916 | 7 | 17 | 3 | 16 | 6.8 | 0 | 212 | 1931 | 3 | 10 | 2 | 56 | 6.1 | 2 |
| 47 | 1897 | 12 | 4 | 9 | 18 | 6.2 | 2 | 130 | 1916 | 8 | 8 | 13 | 25 | 6.3 | 0 | 213 | 1931 | 6 | 23 | 15 | 14 | 6.3 | 1 |
| 48 | 1897 | 12 | 26 | 16 | 41 | 6.2 | 2 | 131 | 1916 | 8 | 21 | 23 | 33 | 6.2 | 2 | 214 | 1931 | 6 | 23 | 15 | 14 | 6.6 | 0 |
| 49 | 1898 | 4 | 23 | 8 | 37 | 7.2 | 0 | 132 | 1916 | 8 | 28 | 7 | 43 | 6.8 | 0 | 215 | 1931 | 8 | 18 | 14 | 40 | 6.0 | 2 |
| 50 | 1898 | 10 | 7 | 11 | 0 | 6.0 | 0 | 133 | 1916 | 11 | 24 | 13 | 4 | 6.6 | 0 | 216 | 1931 | 9 | 9 | 4 | 8 | 6.3 | 2 |
| 51 | 1898 | 12 | 16 | 1 | 47 | 6.0 | 2 | 134 | 1917 | 3 | 15 | 9 | 14 | 6.9 | 0 | 217 | 1932 | 6 | 22 | 9 | 36 | 6.2 | 0 |
| 52 | 1899 | 3 | 22 | 19 | 23 | 6.5 | 2 | 135 | 1917 | 4 | 21 | 12 | 53 | 6.3 | 2 | 218 | 1932 | 6 | 30 | 3 | 16 | 6.0 | 0 |
| 53 | 1899 | 8 | 3 | 18 | 52 | 6.0 | 0 | 136 | 1917 | 6 | 14 | 22 | 22 | 6.1 | 0 | 219 | 1932 | 7 | 10 | 16 | 45 | 6.1 | 0 |
| 54 | 1900 | 3 | 12 | 10 | 34 | 6.4 | 0 | 137 | 1917 | 7 | 29 | 23 | 32 | 7.3 | 0 | 220 | 1932 | 9 | 3 | 20 | 58 | 6.6 | 0 |
| 55 | 1900 | 8 | 5 | 13 | 21 | 6.6 | 0 | 138 | 1917 | 11 | 16 | 0 | 2 | 6.0 | 0 | 221 | 1932 | 9 | 5 | 12 | 8 | 6.1 | 2 |
| 56 | 1900 | 8 | 29 | 11 | 32 | 6.8 | 0 | 139 | 1917 | 12 | 6 | 20 | 39 | 6.3 | 0 | 222 | 1933 | 1 | 4 | 0 | 26 | 6.1 | 1 |
| 57 | 1900 | 9 | 24 | 12 | 32 | 6.0 | 2 | 140 | 1918 | 7 | 26 | 5 | 50 | 6.7 | 0 | 223 | 1933 | 1 | 7 | 13 | 6 | 6.8 | 1 |
| 58 | 1901 | 5 | 14 | 5 | 11 | 6.0 | 0 | 141 | 1918 | 9 | 13 | 18 | 8 | 6.1 | 0 | 224 | 1933 | 1 | 8 | 15 | 28 | 6.3 | 1 |
| 59 | 1901 | 6 | 15 | 18 | 34 | 7.0 | 2 | 142 | 1918 | 12 | 14 | 6 | 33 | 6.2 | 0 | 225 | 1933 | 3 | 3 | 2 | 30 | 8.1 | 0 |
| 60 | 1901 | 8 | 9 | 18 | 23 | 7.2 | 1 | 143 | 1919 | 5 | 3 | 9 | 52 | 7.4 | 0 | 226 | 1933 | 3 | 3 | 2 | 40 | 6.5 | 2 |
| 61 | 1901 | 8 | 10 | 3 | 33 | 7.4 | 0 | 144 | 1919 | 7 | 22 | 8 | 51 | 6.1 | 0 | 227 | 1933 | 3 | 3 | 3 | 25 | 6.6 | 2 |
| 62 | 1901 | 8 | 10 | 5 | 0 | 6.3 | 2 | 145 | 1919 | 8 | 4 | 3 | 8 | 6.7 | 0 | 228 | 1933 | 3 | 3 | 3 | 48 | 6.0 | 2 |
| 63 | 1901 | 8 | 11 | 20 | 31 | 6.0 | 2 | 146 | 1919 | 8 | 8 | 1 | 32 | 6.2 | 0 | 229 | 1933 | 3 | 3 | 5 | 42 | 6.8 | 2 |
| 64 | 1901 | 8 | 29 | 21 | 16 | 6.3 | 2 | 147 | 1919 | 9 | 12 | 23 | 54 | 6.1 | 2 | 230 | 1933 | 3 | 3 | 18 | 12 | 6.5 | 2 |
| 65 | 1901 | 9 | 30 | 19 | 44 | 6.2 | 2 | 148 | 1919 | 10 | 11 | 22 | 17 | 6.3 | 0 | 231 | 1933 | 3 | 3 | 18 | 37 | 6.2 | 2 |
| 66 | 1902 | 1 | 1 | 0 | 20 | 6.1 | 2 | 149 | 1919 | 12 | 20 | 9 | 28 | 6.3 | 0 | 232 | 1933 | 3 | 3 | 18 | 39 | 6.0 | 2 |
| 67 | 1902 | 1 | 31 | 10 | 42 | 6.6 | 1 | 150 | 1920 | 2 | 8 | 0 | 6 | 6.7 | 0 | 233 | 1933 | 3 | 4 | 0 | 2 | 6.0 | 2 |
| 68 | 1902 | 5 | 2 | 20 | 31 | 7.0 | 0 | 151 | 1920 | 9 | 17 | 0 | 8 | 6.5 | 0 | 234 | 1933 | 3 | 8 | 10 | 35 | 6.0 | 2 |
| 69 | 1902 | 7 | 1 | 17 | 19 | 6.3 | 2 | 152 | 1920 | 9 | 21 | 5 | 27 | 6.1 | 0 | 235 | 1933 | 4 | 2 | 7 | 41 | 6.0 | 2 |
| 70 | 1902 | 7 | 8 | 23 | 5 | 6.2 | 2 | 153 | 1920 | 11 | 9 | 2 | 38 | 6.3 | 0 | 236 | 1933 | 4 | 9 | 11 | 46 | 6.6 | 2 |
| 71 | 1902 | 8 | 7 | 18 | 22 | 6.1 | 2 | 154 | 1920 | 12 | 3 | 8 | 39 | 6.0 | 2 | 237 | 1933 | 4 | 9 | 11 | 57 | 6.0 | 2 |
| 72 | 1904 | 12 | 17 | 16 | 2 | 6.3 | 0 | 155 | 1920 | 12 | 20 | 5 | 11 | 6.8 | 0 | 238 | 1933 | 4 | 19 | 11 | 55 | 6.1 | 2 |
| 73 | 1904 | 12 | 24 | 11 | 46 | 6.0 | 0 | 156 | 1921 | 1 | 10 | 3 | 56 | 6.1 | 0 | 239 | 1933 | 4 | 23 | 16 | 13 | 6.3 | 2 |
| 74 | 1905 | 6 | 27 | 1 | 10 | 6.2 | 0 | 157 | 1921 | 3 | 3 | 12 | 2 | 6.9 | 0 | 240 | 1933 | 6 | 9 | 3 | 10 | 6.1 | 2 |
| 75 | 1905 | 7 | 7 | 1 | 21 | 7.8 | 0 | 158 | 1921 | 8 | 22 | 13 | 5 | 6.3 | 0 | 241 | 1933 | 6 | 13 | 6 | 8 | 6.1 | 0 |
| 76 | 1905 | 12 | 26 | 12 | 11 | 6.0 | 0 | 159 | 1922 | 1 | 23 | 7 | 5 | 6.5 | 0 | 242 | 1933 | 6 | 14 | 5 | 33 | 6.4 | 0 |
| 77 | 1906 | 2 | 4 | 15 | 24 | 6.0 | 2 | 160 | 1922 | 3 | 17 | 3 | 31 | 6.2 | 0 | 243 | 1933 | 6 | 19 | 6 | 37 | 7.1 | 0 |
| 78 | 1906 | 4 | 5 | 11 | 50 | 6.0 | 2 | 161 | 1922 | 4 | 27 | 18 | 15 | 6.0 | 2 | 244 | 1933 | 7 | 10 | 9 | 21 | 6.3 | 2 |
| 79 | 1906 | 4 | 9 | 2 | 37 | 6.1 | 0 | 162 | 1922 | 5 | 16 | 5 | 21 | 6.5 | 0 | 245 | 1933 | 7 | 21 | 8 | 14 | 6.2 | 2 |
| 80 | 1907 | 1 | 5 | 1 | 46 | 6.3 | 0 | 163 | 1922 | 6 | 3 | 13 | 56 | 6.1 | 2 | 246 | 1933 | 9 | 21 | 18 | 47 | 6.1 | 2 |
| 81 | 1907 | 4 | 23 | 9 | 57 | 6.0 | 0 | 164 | 1922 | 7 | 6 | 5 | 20 | 6.5 | 0 | 247 | 1934 | 4 | 7 | 4 | 9 | 6.3 | 0 |
| 82 | 1907 | 5 | 23 | 7 | 54 | 6.3 | 0 | 165 | 1922 | 12 | 9 | 7 | 33 | 6.8 | 0 | 248 | 1934 | 10 | 6 | 5 | 25 | 6.2 | 0 |
| 83 | 1907 | 12 | 2 | 22 | 53 | 6.7 | 0 | 166 | 1923 | 5 | 26 | 12 | 12 | 6.4 | 1 | 249 | 1935 | 2 | 20 | 5 | 10 | 6.0 | 0 |

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*Table 1 (continued)*

| NO | YEAR | MO | DY | HR | MN | MAG | C | NO | YEAR | MO | DY | HR | MN | MAG | C | NO | YEAR | MO | DY | HR | MN | MAG | C |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 250 | 1935 | 3 | 31 | 6 | 19 | 6.4 | 0 | 328 | 1943 | 3 | 14 | 21 | 43 | 6.3 | 1 | 406 | 1960 | 3 | 21 | 2 | 7 | 7.2 | 0 |
| 251 | 1935 | 7 | 19 | 9 | 50 | 6.9 | 0 | 329 | 1943 | 4 | 11 | 23 | 46 | 6.7 | 0 | 407 | 1960 | 3 | 21 | 9 | 34 | 6.0 | 2 |
| 252 | 1935 | 10 | 13 | 1 | 45 | 6.9 | 2 | 330 | 1943 | 4 | 13 | 4 | 43 | 6.2 | 2 | 408 | 1960 | 3 | 23 | 9 | 23 | 6.7 | 2 |
| 253 | 1935 | 10 | 13 | 2 | 0 | 6.5 | 2 | 331 | 1943 | 4 | 13 | 4 | 50 | 6.0 | 2 | 409 | 1960 | 3 | 23 | 10 | 7 | 6.1 | 2 |
| 254 | 1935 | 10 | 13 | 10 | 57 | 6.4 | 1 | 332 | 1943 | 6 | 13 | 14 | 11 | 7.1 | 0 | 410 | 1960 | 3 | 24 | 7 | 22 | 6.0 | 2 |
| 255 | 1935 | 10 | 18 | 9 | 11 | 7.1 | 0 | 333 | 1943 | 6 | 13 | 14 | 58 | 6.4 | 2 | 411 | 1960 | 6 | 16 | 0 | 36 | 6.2 | 1 |
| 256 | 1935 | 10 | 18 | 23 | 53 | 6.5 | 2 | 334 | 1943 | 6 | 13 | 17 | 36 | 6.4 | 2 | 412 | 1960 | 7 | 30 | 2 | 31 | 6.7 | 0 |
| 257 | 1935 | 10 | 19 | 6 | 52 | 6.3 | 2 | 335 | 1943 | 6 | 14 | 2 | 39 | 6.3 | 2 | 413 | 1960 | 8 | 13 | 16 | 11 | 6.2 | 0 |
| 258 | 1936 | 3 | 2 | 12 | 19 | 6.8 | 0 | 336 | 1943 | 6 | 15 | 1 | 22 | 6.1 | 2 | 414 | 1961 | 1 | 16 | 16 | 20 | 6.8 | 0 |
| 259 | 1936 | 3 | 11 | 5 | 36 | 6.3 | 2 | 337 | 1943 | 6 | 15 | 20 | 10 | 6.7 | 2 | 415 | 1961 | 1 | 16 | 20 | 19 | 6.4 | 2 |
| 260 | 1936 | 3 | 11 | 9 | 44 | 6.1 | 0 | 338 | 1944 | 2 | 1 | 14 | 16 | 6.8 | 0 | 416 | 1961 | 1 | 16 | 21 | 12 | 6.5 | 2 |
| 261 | 1936 | 6 | 3 | 11 | 15 | 6.0 | 0 | 339 | 1944 | 3 | 10 | 15 | 40 | 6.1 | 0 | 417 | 1961 | 1 | 16 | 23 | 3 | 6.1 | 2 |
| 262 | 1936 | 11 | 3 | 5 | 45 | 7.5 | 0 | 340 | 1944 | 6 | 6 | 20 | 48 | 6.0 | 0 | 418 | 1961 | 1 | 17 | 0 | 41 | 6.6 | 2 |
| 263 | 1936 | 11 | 14 | 9 | 58 | 6.0 | 0 | 341 | 1944 | 10 | 3 | 5 | 29 | 6.4 | 1 | 419 | 1961 | 2 | 23 | 13 | 16 | 6.4 | 0 |
| 264 | 1937 | 1 | 7 | 15 | 12 | 6.6 | 2 | 342 | 1945 | 2 | 10 | 13 | 57 | 7.1 | 0 | 420 | 1961 | 3 | 25 | 7 | 57 | 6.1 | 0 |
| 265 | 1937 | 1 | 20 | 9 | 3 | 6.0 | 0 | 343 | 1945 | 2 | 18 | 7 | 35 | 6.2 | 2 | 421 | 1962 | 4 | 12 | 9 | 52 | 6.8 | 0 |
| 266 | 1937 | 3 | 22 | 4 | 27 | 6.1 | 0 | 344 | 1945 | 2 | 18 | 19 | 8 | 6.6 | 2 | 422 | 1962 | 4 | 26 | 0 | 47 | 6.4 | 2 |
| 267 | 1937 | 7 | 27 | 4 | 56 | 7.1 | 2 | 345 | 1945 | 3 | 12 | 6 | 37 | 6.6 | 0 | 423 | 1963 | 5 | 8 | 19 | 22 | 6.1 | 0 |
| 268 | 1937 | 10 | 17 | 13 | 47 | 6.6 | 0 | 346 | 1945 | 4 | 10 | 10 | 22 | 6.4 | 0 | 424 | 1963 | 8 | 15 | 15 | 11 | 6.6 | 0 |
| 269 | 1937 | 12 | 10 | 22 | 28 | 6.1 | 0 | 347 | 1945 | 6 | 26 | 8 | 40 | 6.1 | 0 | 425 | 1964 | 1 | 10 | 13 | 50 | 6.1 | 0 |
| 270 | 1938 | 5 | 23 | 16 | 18 | 7.0 | 0 | 348 | 1946 | 5 | 10 | 7 | 27 | 6.1 | 0 | 426 | 1964 | 2 | 5 | 20 | 30 | 6.0 | 0 |
| 271 | 1938 | 9 | 22 | 3 | 52 | 6.5 | 0 | 349 | 1946 | 7 | 20 | 6 | 16 | 6.0 | 0 | 427 | 1964 | 4 | 16 | 10 | 4 | 6.0 | 0 |
| 272 | 1938 | 10 | 12 | 9 | 34 | 6.9 | 0 | 350 | 1946 | 8 | 3 | 22 | 6 | 6.1 | 0 | 428 | 1964 | 5 | 30 | 23 | 30 | 6.2 | 0 |
| 273 | 1938 | 10 | 29 | 22 | 8 | 6.4 | 0 | 351 | 1946 | 8 | 14 | 18 | 40 | 6.0 | 0 | 429 | 1965 | 3 | 17 | 1 | 46 | 6.4 | 0 |
| 274 | 1938 | 11 | 5 | 17 | 43 | 7.5 | 0 | 352 | 1947 | 1 | 3 | 12 | 57 | 6.0 | 0 | 430 | 1965 | 3 | 29 | 19 | 47 | 6.4 | 2 |
| 275 | 1938 | 11 | 5 | 19 | 50 | 7.3 | 2 | 353 | 1947 | 11 | 14 | 19 | 49 | 6.5 | 0 | 431 | 1965 | 6 | 13 | 16 | 6 | 6.0 | 0 |
| 276 | 1938 | 11 | 6 | 6 | 22 | 6.1 | 2 | 354 | 1948 | 3 | 15 | 20 | 24 | 6.0 | 0 | 432 | 1965 | 9 | 18 | 1 | 21 | 6.7 | 0 |
| 277 | 1938 | 11 | 6 | 17 | 53 | 7.4 | 2 | 355 | 1948 | 5 | 12 | 9 | 57 | 6.6 | 0 | 433 | 1965 | 9 | 23 | 7 | 8 | 6.2 | 2 |
| 278 | 1938 | 11 | 7 | 2 | 19 | 6.0 | 2 | 356 | 1948 | 5 | 12 | 10 | 21 | 6.1 | 2 | 434 | 1967 | 1 | 17 | 20 | 59 | 6.3 | 0 |
| 279 | 1938 | 11 | 7 | 6 | 4 | 6.2 | 2 | 357 | 1948 | 5 | 14 | 22 | 19 | 6.2 | 2 | 435 | 1967 | 11 | 19 | 21 | 6 | 6.0 | 0 |
| 280 | 1938 | 11 | 7 | 6 | 38 | 6.9 | 2 | 358 | 1948 | 9 | 23 | 9 | 52 | 6.0 | 0 | 436 | 1968 | 5 | 16 | 9 | 48 | 7.9 | 0 |
| 281 | 1938 | 11 | 7 | 9 | 48 | 6.2 | 2 | 359 | 1948 | 10 | 29 | 5 | 45 | 6.3 | 0 | 437 | 1968 | 5 | 16 | 10 | 4 | 6.2 | 2 |
| 282 | 1938 | 11 | 7 | 10 | 38 | 6.4 | 2 | 360 | 1949 | 5 | 22 | 6 | 40 | 6.3 | 0 | 438 | 1968 | 5 | 16 | 19 | 39 | 7.5 | 2 |
| 283 | 1938 | 11 | 7 | 10 | 45 | 6.2 | 2 | 361 | 1951 | 7 | 26 | 18 | 59 | 6.0 | 0 | 439 | 1968 | 5 | 17 | 1 | 13 | 6.1 | 2 |
| 284 | 1938 | 11 | 7 | 10 | 54 | 6.4 | 2 | 362 | 1951 | 7 | 29 | 8 | 4 | 6.1 | 0 | 440 | 1968 | 5 | 17 | 8 | 4 | 6.7 | 2 |
| 285 | 1938 | 11 | 7 | 11 | 27 | 6.0 | 2 | 363 | 1951 | 10 | 18 | 17 | 26 | 6.6 | 0 | 441 | 1968 | 5 | 23 | 4 | 29 | 6.3 | 2 |
| 286 | 1938 | 11 | 7 | 13 | 15 | 6.3 | 2 | 364 | 1952 | 3 | 4 | 10 | 22 | 8.2 | 0 | 442 | 1968 | 5 | 24 | 23 | 6 | 6.2 | 2 |
| 287 | 1938 | 11 | 8 | 4 | 33 | 6.5 | 2 | 365 | 1952 | 3 | 4 | 10 | 40 | 6.5 | 2 | 443 | 1968 | 6 | 12 | 22 | 41 | 7.2 | 2 |
| 288 | 1938 | 11 | 9 | 11 | 22 | 6.5 | 2 | 366 | 1952 | 3 | 5 | 4 | 56 | 6.4 | 2 | 444 | 1968 | 6 | 13 | 6 | 57 | 6.1 | 2 |
| 289 | 1938 | 11 | 9 | 18 | 15 | 6.3 | 2 | 367 | 1952 | 3 | 5 | 18 | 17 | 6.0 | 2 | 445 | 1968 | 6 | 17 | 20 | 52 | 6.4 | 2 |
| 290 | 1938 | 11 | 11 | 7 | 22 | 6.4 | 2 | 368 | 1952 | 3 | 10 | 2 | 3 | 6.8 | 2 | 446 | 1968 | 6 | 18 | 3 | 57 | 6.0 | 0 |
| 291 | 1938 | 11 | 14 | 7 | 31 | 7.0 | 2 | 369 | 1952 | 4 | 28 | 19 | 54 | 6.2 | 2 | 447 | 1968 | 6 | 22 | 10 | 12 | 6.1 | 2 |
| 292 | 1938 | 11 | 14 | 11 | 36 | 6.0 | 2 | 370 | 1952 | 5 | 14 | 9 | 36 | 6.1 | 2 | 448 | 1968 | 7 | 5 | 20 | 28 | 6.4 | 0 |
| 293 | 1938 | 11 | 16 | 20 | 8 | 6.6 | 2 | 371 | 1952 | 5 | 17 | 18 | 48 | 6.1 | 2 | 449 | 1968 | 7 | 12 | 9 | 44 | 6.4 | 2 |
| 294 | 1938 | 11 | 19 | 14 | 54 | 6.0 | 2 | 372 | 1952 | 5 | 20 | 3 | 32 | 6.5 | 2 | 450 | 1968 | 9 | 21 | 22 | 6 | 6.9 | 2 |
| 295 | 1938 | 11 | 22 | 10 | 14 | 6.9 | 2 | 373 | 1952 | 10 | 27 | 0 | 46 | 6.1 | 1 | 451 | 1968 | 10 | 8 | 5 | 49 | 6.2 | 2 |
| 296 | 1938 | 11 | 22 | 10 | 40 | 6.1 | 2 | 374 | 1952 | 10 | 27 | 0 | 53 | 6.4 | 1 | 452 | 1968 | 11 | 11 | 23 | 41 | 6.0 | 2 |
| 297 | 1938 | 11 | 22 | 12 | 24 | 6.0 | 2 | 375 | 1952 | 10 | 27 | 3 | 1 | 6.4 | 1 | 453 | 1968 | 11 | 14 | 3 | 41 | 6.0 | 2 |
| 298 | 1938 | 11 | 25 | 17 | 20 | 6.3 | 2 | 376 | 1952 | 10 | 27 | 4 | 19 | 6.5 | 0 | 454 | 1968 | 11 | 25 | 6 | 20 | 6.0 | 2 |
| 299 | 1938 | 11 | 29 | 22 | 39 | 6.4 | 2 | 377 | 1952 | 10 | 27 | 12 | 17 | 6.4 | 2 | 455 | 1969 | 3 | 17 | 0 | 54 | 6.1 | 0 |
| 300 | 1938 | 11 | 30 | 11 | 29 | 6.9 | 2 | 378 | 1952 | 10 | 28 | 15 | 30 | 6.3 | 2 | 456 | 1970 | 5 | 28 | 4 | 5 | 6.2 | 0 |
| 301 | 1938 | 12 | 1 | 0 | 16 | 6.1 | 2 | 379 | 1952 | 11 | 1 | 1 | 37 | 6.4 | 2 | 457 | 1970 | 5 | 28 | 7 | 35 | 6.0 | 2 |
| 302 | 1938 | 12 | 3 | 21 | 12 | 6.5 | 2 | 380 | 1953 | 1 | 19 | 13 | 57 | 6.0 | 2 | 458 | 1970 | 9 | 14 | 18 | 44 | 6.2 | 0 |
| 303 | 1938 | 12 | 7 | 22 | 4 | 6.4 | 0 | 381 | 1953 | 2 | 6 | 22 | 13 | 6.7 | 2 | 459 | 1970 | 12 | 7 | 5 | 20 | 6.1 | 0 |
| 304 | 1938 | 12 | 14 | 2 | 26 | 6.3 | 2 | 382 | 1953 | 4 | 4 | 14 | 52 | 6.2 | 0 | 460 | 1971 | 4 | 5 | 3 | 39 | 6.0 | 0 |
| 305 | 1938 | 12 | 19 | 6 | 45 | 6.0 | 2 | 383 | 1953 | 5 | 26 | 10 | 43 | 6.0 | 2 | 461 | 1971 | 8 | 2 | 16 | 24 | 7.0 | 0 |
| 306 | 1938 | 12 | 23 | 10 | 51 | 6.0 | 2 | 384 | 1953 | 12 | 7 | 23 | 11 | 6.4 | 0 | 462 | 1971 | 9 | 15 | 23 | 55 | 6.3 | 0 |
| 307 | 1939 | 1 | 24 | 13 | 1 | 6.1 | 2 | 385 | 1953 | 12 | 22 | 2 | 36 | 6.1 | 0 | 463 | 1971 | 9 | 24 | 10 | 9 | 6.1 | 2 |
| 308 | 1939 | 2 | 17 | 3 | 51 | 6.5 | 2 | 386 | 1954 | 4 | 5 | 8 | 14 | 6.1 | 2 | 464 | 1972 | 3 | 20 | 0 | 57 | 6.4 | 0 |
| 309 | 1939 | 8 | 22 | 9 | 6 | 6.3 | 2 | 387 | 1954 | 7 | 18 | 18 | 7 | 6.4 | 0 | 465 | 1973 | 9 | 5 | 22 | 3 | 6.1 | 0 |
| 310 | 1939 | 10 | 11 | 3 | 32 | 7.0 | 0 | 388 | 1954 | 9 | 12 | 16 | 43 | 6.2 | 0 | 466 | 1973 | 9 | 10 | 3 | 25 | 6.0 | 2 |
| 311 | 1939 | 10 | 11 | 3 | 51 | 6.4 | 2 | 389 | 1954 | 11 | 19 | 5 | 44 | 6.1 | 0 | 467 | 1973 | 11 | 19 | 22 | 1 | 6.4 | 0 |
| 312 | 1940 | 2 | 9 | 22 | 53 | 6.2 | 0 | 390 | 1955 | 5 | 1 | 18 | 55 | 6.1 | 0 | 468 | 1974 | 1 | 25 | 4 | 12 | 6.0 | 0 |
| 313 | 1940 | 11 | 14 | 19 | 33 | 6.2 | 0 | 391 | 1955 | 5 | 6 | 9 | 4 | 6.0 | 2 | 469 | 1974 | 3 | 3 | 13 | 50 | 6.1 | 0 |
| 314 | 1940 | 11 | 20 | 0 | 1 | 6.6 | 0 | 392 | 1956 | 2 | 10 | 9 | 2 | 6.0 | 0 | 470 | 1974 | 7 | 8 | 14 | 45 | 6.3 | 0 |
| 315 | 1941 | 2 | 9 | 13 | 16 | 6.1 | 0 | 393 | 1956 | 10 | 12 | 21 | 22 | 6.1 | 0 | 471 | 1974 | 10 | 10 | 15 | 48 | 6.2 | 1 |
| 316 | 1941 | 3 | 12 | 23 | 16 | 6.3 | 0 | 394 | 1956 | 11 | 21 | 16 | 33 | 6.2 | 0 | 472 | 1974 | 10 | 10 | 15 | 56 | 6.4 | 0 |
| 317 | 1941 | 3 | 13 | 6 | 37 | 6.2 | 2 | 395 | 1957 | 6 | 12 | 17 | 28 | 6.1 | 0 | 473 | 1974 | 10 | 12 | 15 | 14 | 6.2 | 2 |
| 318 | 1941 | 3 | 14 | 23 | 31 | 6.2 | 2 | 396 | 1958 | 2 | 16 | 15 | 4 | 6.1 | 0 | 474 | 1974 | 11 | 16 | 8 | 32 | 6.1 | 0 |
| 319 | 1941 | 3 | 19 | 11 | 45 | 6.3 | 2 | 397 | 1958 | 4 | 8 | 3 | 5 | 6.7 | 0 | 475 | 1975 | 5 | 4 | 18 | 31 | 6.0 | 0 |
| 320 | 1941 | 5 | 9 | 18 | 32 | 6.1 | 0 | 398 | 1958 | 4 | 8 | 3 | 30 | 6.2 | 2 | 476 | 1975 | 10 | 30 | 10 | 41 | 6.0 | 0 |
| 321 | 1941 | 5 | 9 | 18 | 38 | 6.0 | 2 | 399 | 1958 | 4 | 8 | 3 | 38 | 6.2 | 2 | 477 | 1976 | 11 | 8 | 17 | 19 | 6.2 | 0 |
| 322 | 1941 | 11 | 26 | 0 | 20 | 6.3 | 0 | 400 | 1958 | 4 | 11 | 9 | 58 | 6.3 | 2 | 478 | 1978 | 2 | 20 | 13 | 36 | 6.7 | 0 |
| 323 | 1942 | 2 | 21 | 16 | 7 | 6.5 | 0 | 401 | 1958 | 9 | 3 | 17 | 10 | 6.0 | 0 | 479 | 1978 | 4 | 7 | 8 | 29 | 6.1 | 0 |
| 324 | 1942 | 9 | 9 | 1 | 7 | 6.3 | 1 | 402 | 1959 | 1 | 22 | 14 | 10 | 6.8 | 0 | 480 | 1978 | 6 | 12 | 17 | 14 | 7.4 | 0 |
| 325 | 1942 | 9 | 21 | 14 | 53 | 6.2 | 0 | 403 | 1959 | 1 | 24 | 14 | 8 | 6.2 | 2 | 481 | 1978 | 6 | 14 | 20 | 34 | 6.3 | 2 |
| 326 | 1942 | 11 | 16 | 2 | 12 | 6.5 | 0 | 404 | 1959 | 10 | 26 | 16 | 35 | 6.8 | 0 | 482 | 1979 | 2 | 20 | 15 | 32 | 6.5 | 0 |
| 327 | 1943 | 3 | 14 | 20 | 59 | 6.1 | 1 | 405 | 1960 | 2 | 5 | 1 | 50 | 6.1 | 0 | 483 | 1980 | 1 | 13 | 0 | 57 | 6.1 | 0 |

NOTE: NO, YEAR, MO, DY, HR, MN, and MAG, indicate earthquake number, year, month, day, hour, minute, and magnitude, respectively. In the column C, 0, 1, and 2 represent main shock, foreshock, and aftershock, respectively.

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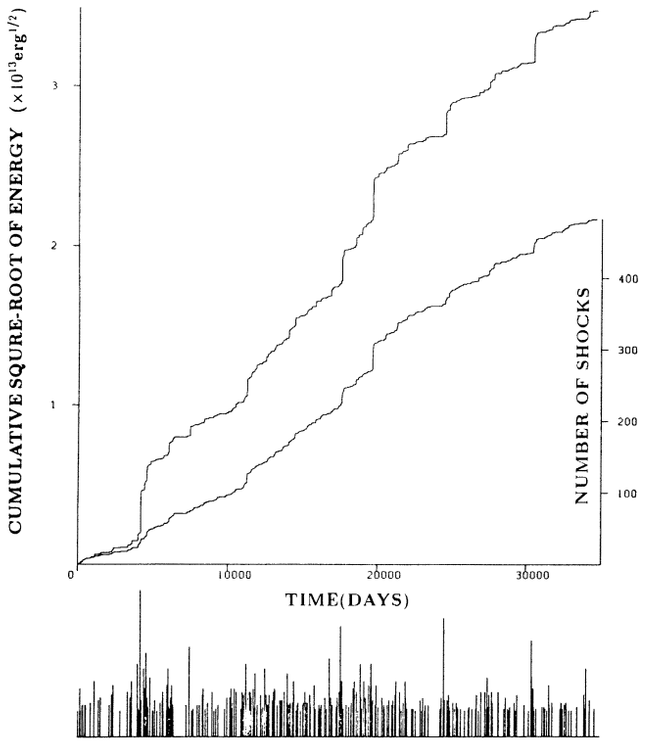



*Figure 2. Cumulative Number of Shocks (lower solid line), Cumulative Square Root of the Released Energy (upper solid line), and Plot of the Magnitudes Versus the Occurrence Times of Shocks.*

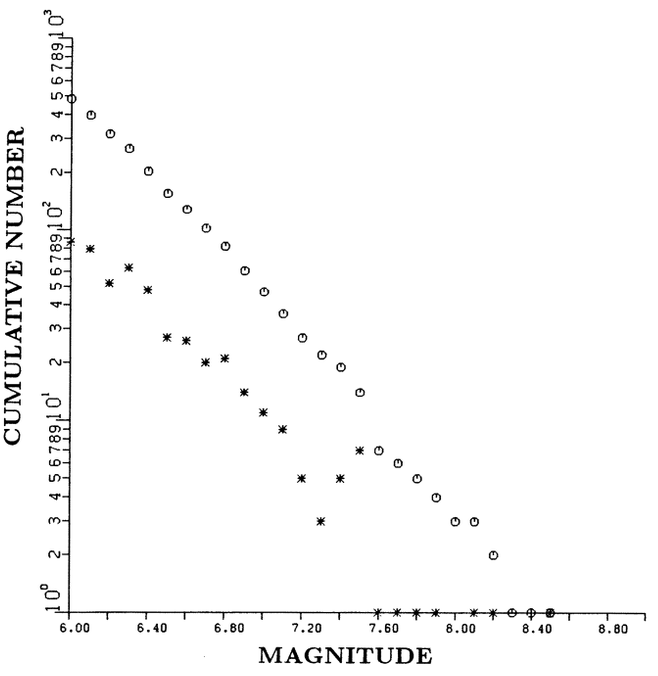



*Figure 3. Density Distribution (*) and Cumulative Distribution (○) of Magnitudes for the Data Set.*

the parameter vectors $\eta$ and $\theta$ have no common components. Then maximization of the full likelihood is reduced to the maximization of each component. Although I will have to examine the first part of (22) afterwards in Section 4 to argue for the utility of seismic quiescence, here I am interested in the second part (the so-called conditional likelihood), the logarithm of which is written as

$$\log L(\theta) = \sum_{i=1}^{N} \log \lambda(t_i; \theta) - \int_0^T \lambda(t; \theta)\, dt, \quad (23)$$

where $\lambda(t; \theta)$ is the parameterized conditional intensity rate discussed in the previous section and $\{t_i\}$ is the set of occurrence times of earthquakes in an observed time interval $[0, T]$. Note here that the magnitude data $\{m_i\}$ of the shocks are also included in the function $\lambda(t; \theta)$, as in (17) and (21). We can expect the conditional likelihood to enjoy the standard large-sample theory under some regularity conditions such as stationarity and ergodicity of the joint series $\{(t_i, m_i)\}$.

The maximum of the log-likelihood and the estimates of the parameters in (23) can be obtained numerically by a standard nonlinear optimization technique such as that in Fletcher and Powell (1963). It is then possible to judge which of the models described in the preceding section provides the best fit to the earthquake process data. For this purpose, the Akaike information criterion (AIC) (Akaike 1974) is used as a measure for selecting the best among competing models for a fixed data set. This is defined by AIC = $(-2)$max(log-likelihood) + 2 (number of used parameters). The model with the smaller AIC shows the better fit to the data. It is sometimes useful to note that the log-likelihood ratio statistic takes the form

$$(-2)\log(L_0/L_1) = \mathrm{AIC}(H_0) - \mathrm{AIC}(H_1) + 2k, \quad (24)$$

where $k$ denotes the difference between the numbers of parameters in $H_0$ and $H_1$. When the model $H_1$ contains the model $H_0$ as a restricted case, then under the null

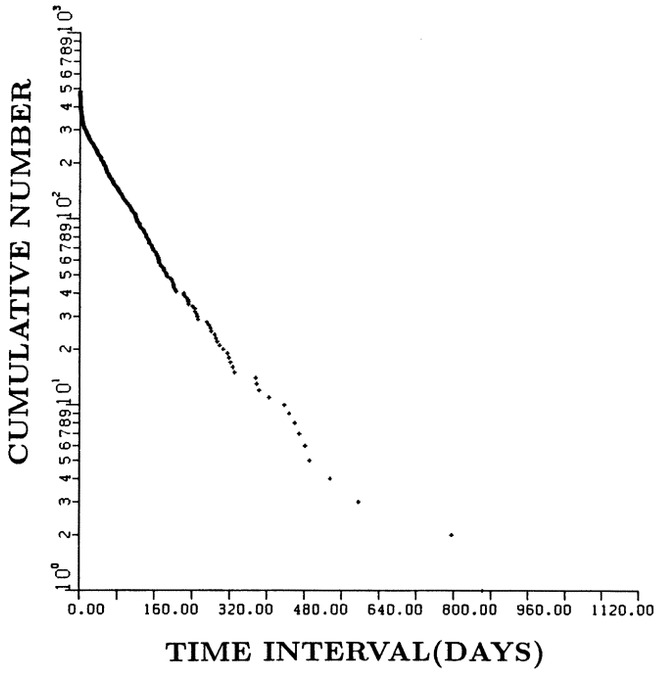



*Figure 4. Empirical Log-Survivor Function of the Intervals Between Successive Earthquakes.*

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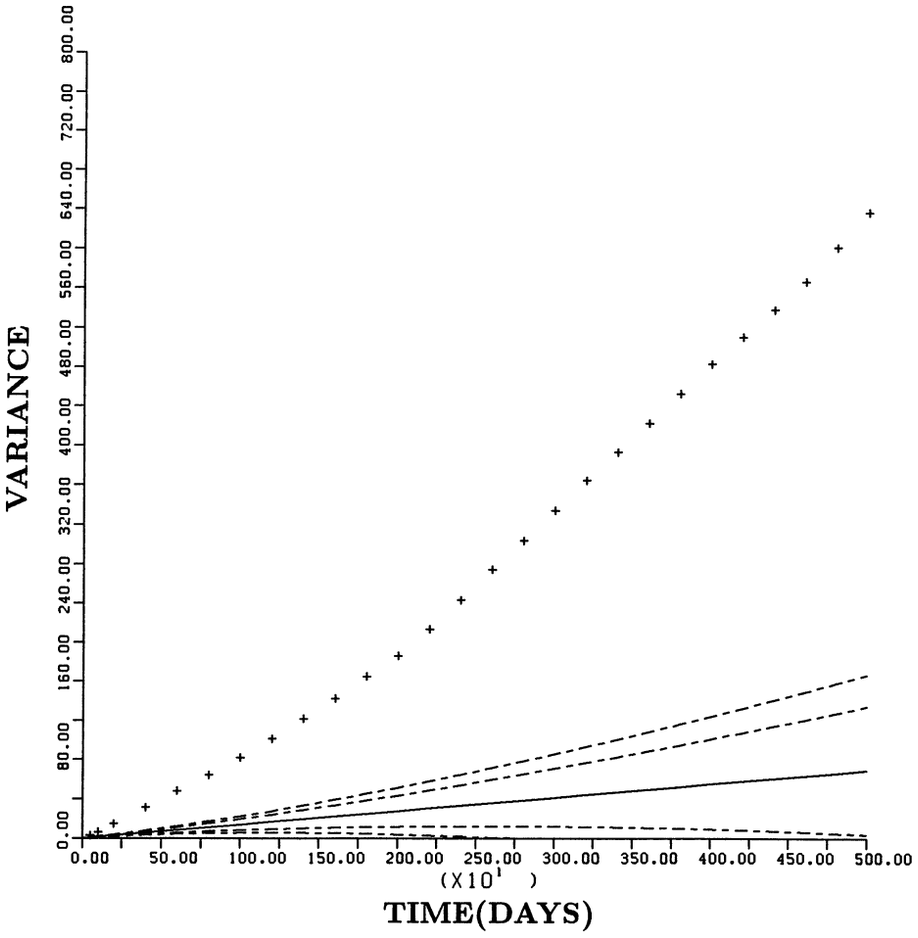



*Figure 5. Estimated Variance–Time Curve (+ + +). The solid line is a theoretical one for the stationary Poisson case, and the dotted lines correspond to the 95.5% and 99.7% error bounds for individual estimates under the stationary Poisson process.*

hypothesis $H_0$, the statistic $(-2)\log(L_0/L_1)$ is expected to have a chi-squared distribution with $k$ df. The comparison of the minimum AIC procedure with the conventional likelihood ratio test is discussed in Akaike (1977, 1983). It is emphasized that we can further extend the comparison among nonnested models by using AIC, which was originally derived for such a purpose.

Table 2 gives the maximum log-likelihoods and AIC values for the models considered. For the models where (15) is used, I considered values of $\kappa$ up to 16, but here are listed only the results for $\kappa = 1$ and for the order $\hat{\kappa}$ with minimum AIC value. Table 2 clearly shows the following: (a) the effect of magnitude on cluster size is significant, (b) for $g(t)$, the modified Omori function (14) is significantly better than the Laguerre-type polynomials (15), and (c) the epidemic-type models (17) are signifi-

*Table 2. Comparison of Models Fitted to the Data in Table 1*

| | *Response function (15), including (13)* | | | | *Response function (11)* | | | |
|---|---|---|---|---|---|---|---|---|
| | *Magnitude effect $\beta = 0$* | | *Magnitude effect $\beta > 0$* | | *Magnitude effect $\beta = 0$* | | *Magnitude effect $\beta > 0$* | |
| *Model* | *$\kappa = 1$** | *$1 \leqq \kappa \leqq 16$** | *$\kappa = 1$** | *$1 \leqq \kappa \leqq 16$** | *$p = 1.0$** | *$p \neq 1.0$** | *$p = 1.0$** | *$p \neq 1.0$** |
| | | | | *Trigger models* | | | | |
| Optimum orders | — | $x = 9$ | — | $x = 7$ | — | — | — | — |
| $-\log L(\theta)$ | 2,404.2 | 2,275.4 | 2,308.6 | 2,271.0 | 2,348.5 | 2,347.3 | 2,251.3 | 2,249.4 |
| Number of parameters | 3 | 11 | 4 | 10 | 3 | 4 | 4 | 5 |
| AIC | 4,814.4 | 4,572.8 | 4,625.3 | 4,562.0 | 4,703.0 | 4,702.6 | 4,510.5 | 4,508.8 |
| | | | | *Epidemic-type models* | | | | |
| Optimum orders | — | $x = 11$ | — | $x = 11$ | — | — | — | — |
| $-\log L(\theta)$ | 2,288.4 | 2,236.1 | 2,248.0 | 2,205.1 | 2,226.4 | 2,226.4 | 2,185.2 | 2,185.0 |
| Number of parameters | 3 | 13 | 4 | 14 | 3 | 4 | 4 | 5 |
| AIC | 4,582.8 | 4,498.3 | 4,504.0 | 4,438.2 | 4,458.8 | 4,460.7 | 4,378.4 | 4,380.0 |

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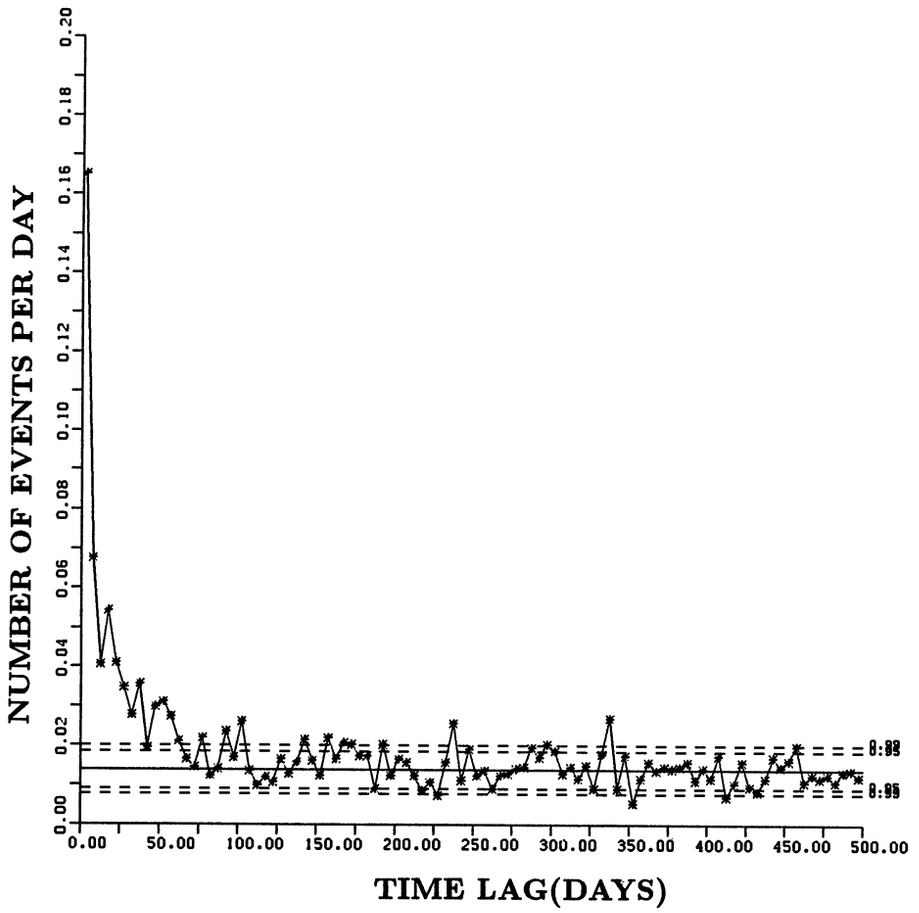



*Figure 6. Histogram for Estimating $m_t(s)$, the Intensity Rate Under the Palm Probability. The dotted lines indicate 95.5% and 99.7% error bounds for individual estimates assuming the stationary Poisson process.*

cantly better than the restricted trigger models (21). Especially, the model (17) with (14) is significantly better than any of the other models. Estimates of the parameters of the best model are listed in Table 3; here the reference magnitude was taken to be $M_r = 6.0$. The plot of the estimated conditional intensity rate is shown in Figure 8.

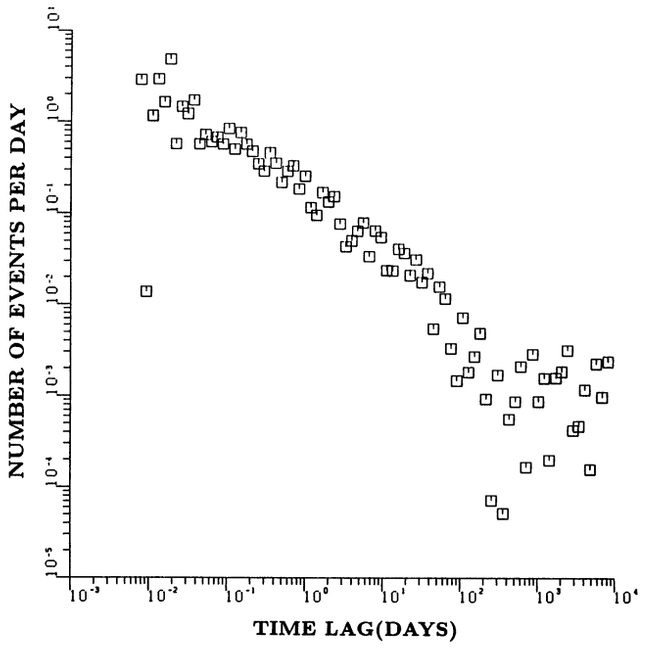



*Figure 7. Plot of $|m_t(s) - \lambda_0|$ Versus s on a Log–Log Scale.*

### 3.3 Residual Analysis of Point Process Data

AIC is useful for the comparison of competing models. Having obtained the best model among those considered, however, there remains the possibility of the existence of a still better model. So, we usually check whether the major features of the given data can be reproduced by the estimated models [see Ogata (1981) for efficient and systematic simulation of point processes using the conditional intensity function]. If any one of the important features is not reproduced, we must consider further models whose AIC values can be compared with those of the previous best model. Thus it may be useful to develop graphical techniques for amplifying the features of the data that deviate from the model, if any.

Suppose that the point process data $\{t_i\}$ are generated

*Table 3. Estimated Parameters of the Best Model ($M_r = 6.0$)*

| $\mu$ | $K$ | $c$ | $p$ | $\beta$ |
|---|---|---|---|---|
| .00536 (shocks/day) | .017284 (shocks/day) | .01959 (days) | 1.0 | 1.61385 |

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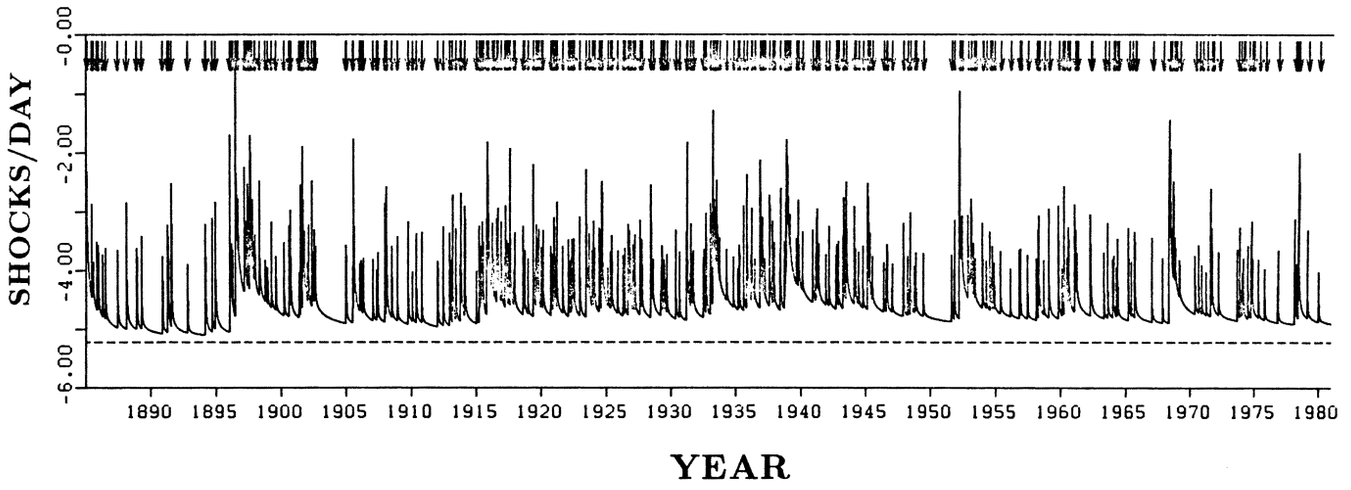



*Figure 8. Estimated Conditional Intensity Rate of (17) With Parameter Values in Table 3. The rate is plotted in the logarithmic scale. The arrows downwards indicate the occurrence times of shocks in the Off Tohoku area.*

by the conditional intensity $\lambda(t)$. Consider the integral of the conditional intensity

$$\Lambda(t) = \int_0^t \lambda(t)\, dt, \tag{25}$$

which is a monotonically increasing function because $\lambda(t)$ is nonnegative. If we consider the random time change $\tau = \Lambda(t)$ from $t$ to $\tau$, then $\{t_i\}$ is transformed one-to-one into $\{\tau_i\}$. It is well known that $\{\tau_i\}$ has the distribution of a stationary Poisson process of intensity 1 (see, e.g., Papangelou 1972). Therefore, if the estimated conditional intensity $\lambda(t; \hat{\theta})$ is a good approximation to the true $\lambda(t)$, then the transformed data $\{\tau_i\}$ are expected to behave like a stationary Poisson process. In other words, a deviation from a property of $\{\tau_i\}$ from that expected of a stationary Poisson process implies the existence of a corresponding feature of the data $\{t_i\}$ that is not captured by the model $\lambda(t; \hat{\theta})$. The intensity $\lambda(t; \hat{\theta})$ represents a model for prediction, whereas the transformed data $\{\tau_i\}$ may be regarded as "noise," or "residuals" in a wide sense, of the point process data $\{t_i\}$. This sequence $\{\tau_i\}$ will be called the *residual process* and is a further example of generalized residuals similar to those discussed in Cox and Snell (1968).

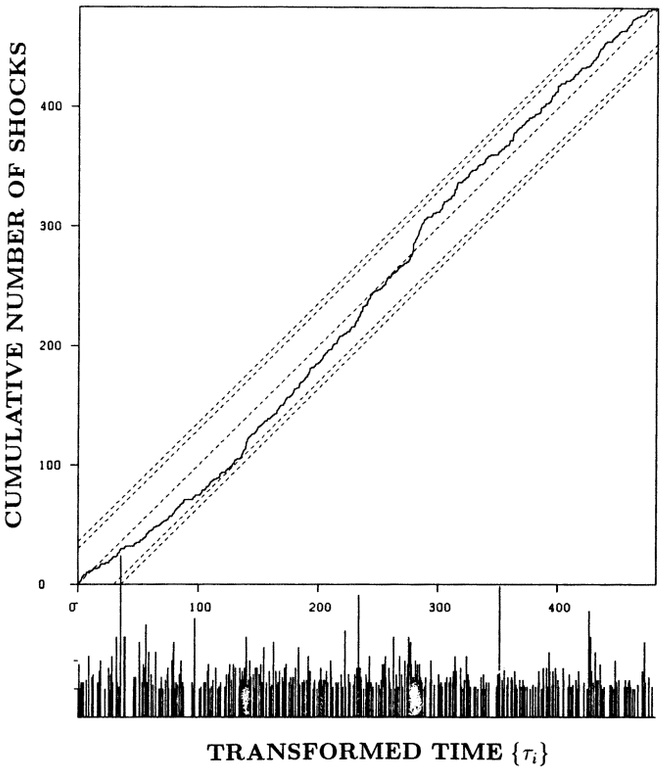



*Figure 9. Cumulative Numbers of the Residual Process and Magnitudes Versus the Transformed Time $\tau_i$. The dotted lines indicate the average and two-sided 95% and 99% error bounds of the Kolmogorov–Smirnov statistic.*

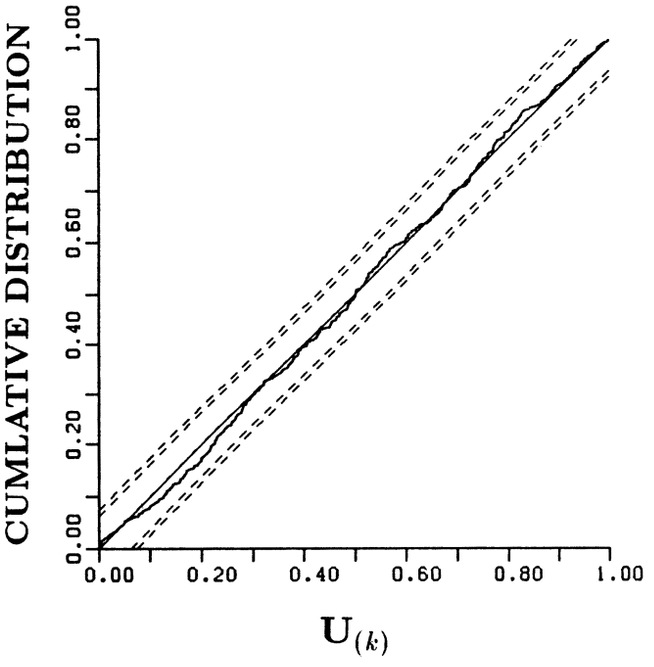



*Figure 10. Empirical Distribution of $U_k$. The dotted lines indicate the 95% and 99% error bounds of the Kolmogorov–Smirnov statistic assuming the uniform distribution.*

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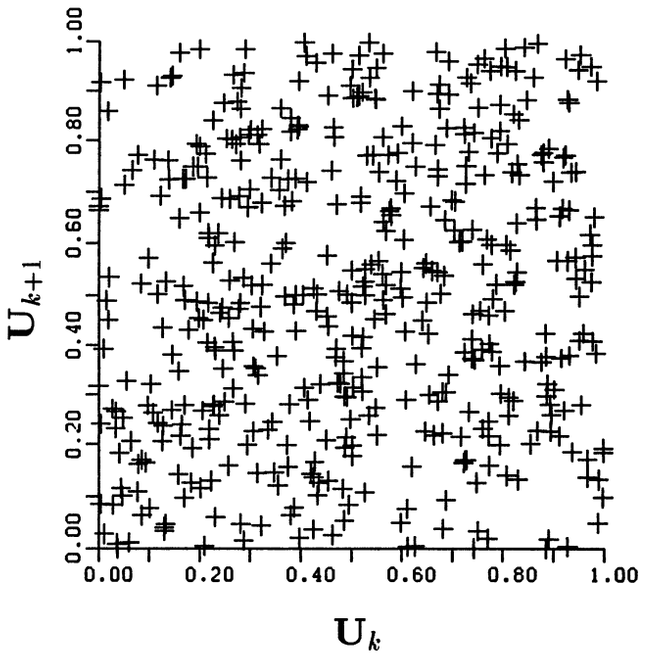



*Figure 11. $U_k$ Versus $U_{k+1}$ Plot.*

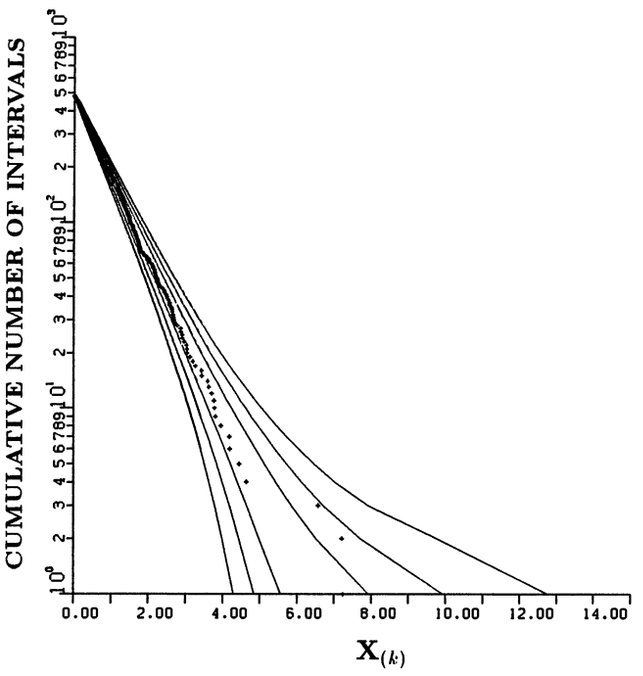



*Figure 12. Empirical Log-Survivor Function of the Intervals $Y_k$ of the Residual Process. The real lines indicate the percentiles that correspond to one-, two-, and three-fold standard errors of the normal distribution.*

Any conventional graphic tests for complete randomness can be useful for residual analysis. Figure 9 shows the plot of the cumulative number of points $\{\tau_i\}$ versus transformed time $\tau = \Lambda(t)$. The dotted lines display the two-sided 95% and 99% error bounds of the Kolmogorov–Smirnov statistics, assuming the uniform empirical distribution. Ogata and Shimazaki (1984) applied this sort of residual analysis of trend to an aftershock sequence to reveal the time at which the background seismicity surpasses the aftershock activity. Another possible test was described by Berman (1983), who discussed the residual analysis of the transformed interarrival times properties; that is to say, he considered whether

$$Y_k = \tau_k - \tau_{k-1} = \Lambda(t_k) - \Lambda(t_{k-1}), \quad k = 1, \ldots, N, \quad (26)$$

are iid exponential random variables with unit mean, and hence whether the statistics $U_k = 1 - \exp(-Y_k)$ are iid uniform random variables on [0, 1). The empirical distri-

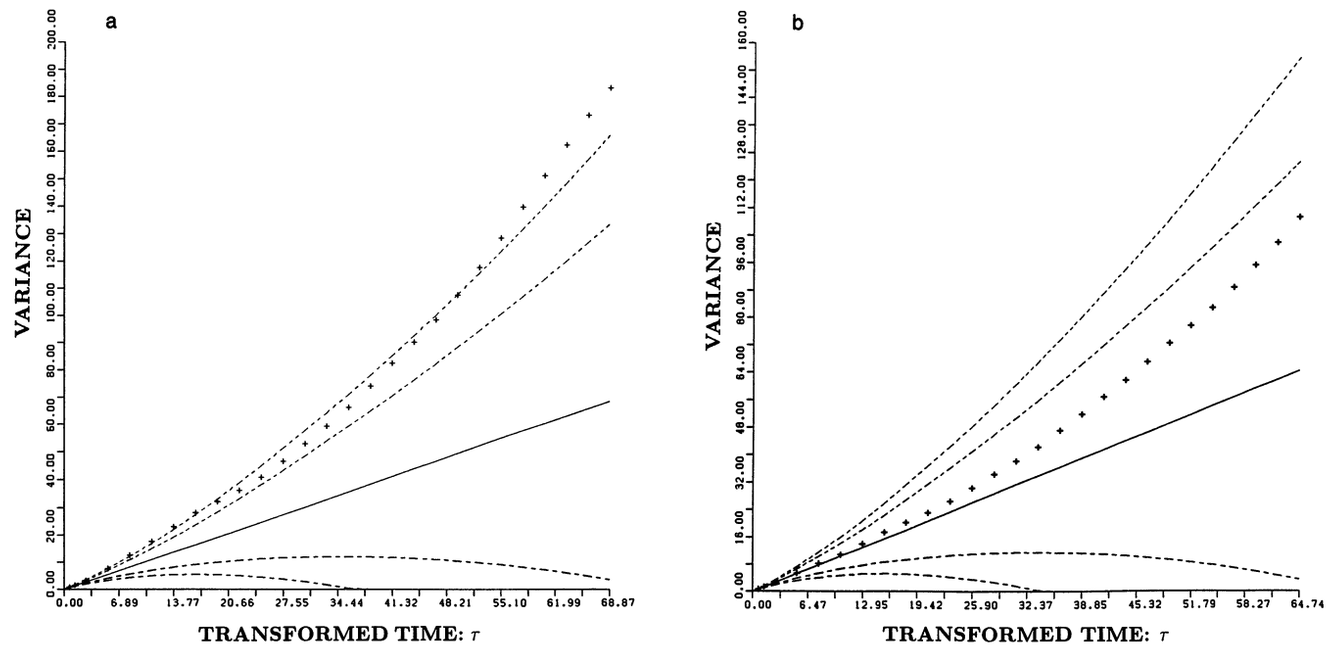



*Figure 13. (a) Estimated Variance–Time Curve for the Residual Process. The solid line is the theoretically expected variance–time curve for the stationary Poisson process. The dotted lines indicate for individual estimates 95.5% and 99.7% error bounds for individual estimates. (b) Estimated variance–time curve for the residual process after removing the Shioya-Oki swarms in 1938.*

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bution of $U_k$ for our data set is plotted in Figure 10 with the same error bounds as in Figure 9. Berman also suggested a plot to test for independence of the intervals. Assuming that any serial correlation present is likely to show up in neighboring intervals, he plotted $U_k$ against $U_{k+1}$. The plot for our data set is given in Figure 11. The plots for $\{\tau_k\}$ corresponding to Figures 4, 5, and 6 are given in Figures 12, 13a, and 14, respectively. Here the lines for two-sided error bounds for individual estimates correspond to 99.7%, 95.5%, and 68.3% (only for Fig. 12). The variance–time curve $V(\tau)$ in Figure 13a still indicates some clustering features, despite the fact that Figures 9–12 and 14 seem to support the hypothesis that the noise process $\{\tau_i\}$ is a stationary Poisson process.

Thus I suspect that there may be some local unusual characteristics in $\{\tau_i\}$ invalidating the stationary Poisson assumption. To find such features, consider the number of points $\Delta N = N(\tau - h, \tau)$ in the interval $(\tau - h, \tau)$. If the residual process $\{\tau_i\}$ is stationary Poisson, then $\Delta N = N(\tau - h, \tau)$ is a Poisson random variable with mean $h$ for each $\tau$. Here, if we use the transformation suggested in Shimizu and Yuasa (1984),

$$\xi = \xi(\Delta N, h) = \frac{33\Delta N + 29 - h - (32\Delta N + 31)[h/(\Delta N + 1)]^{1/4}}{9(\Delta N + 1)^{1/2}}, \tag{27}$$

then $\xi$ is well approximated by a normal random variable with mean 0 and variance 1. This transformation was obtained by an idea similar to the derivation of the well-known Wilson–Hilferty transform.

Setting $h = 8$, for example, the time series of $\xi$ as a function of $\tau$ is plotted in Figure 15(3a). It behaves like a Gaussian process, except for the part of the trajectory around 1938. This is a stretch of *outliers* associated with a swarm of large earthquakes, including magnitudes 7.4 (May 23), 7.7 (Nov. 5), 7.8 (Nov. 5), 7.7 (Nov. 6), and 7.1 (Nov. 6) at Shioya-Oki (Off Fukushima Prefecture) in the southern part of the region shown in Figure 1. After finding these outliers, I learned that Abe (1977) had already reported and described the unusual features of these shocks in detail. That is to say, it is very rare that a swarm includes a series of shocks with such large magnitudes or such a large energy release. In addition, he pointed out the fact that there had been no such major earthquake in the focal region for at least 800 years, whereas the frequency of large earthquakes, $M \geq 7.5$, in other parts of the Off Tohoku area was about one per 100 years. Finally, I considered another variance–time curve $V(\tau)$ in Figure 13b, obtained by removing the swarm from the residuals $\{\tau_i\}$: the corresponding numbers $i$ of the removed data are shown from 274 through 302 in Table 1. This variance–time curve suggests that the remaining part of the residuals satisfies the hypothesis of stationary Poisson process. Therefore, I conclude that the seismic activity (in the sense

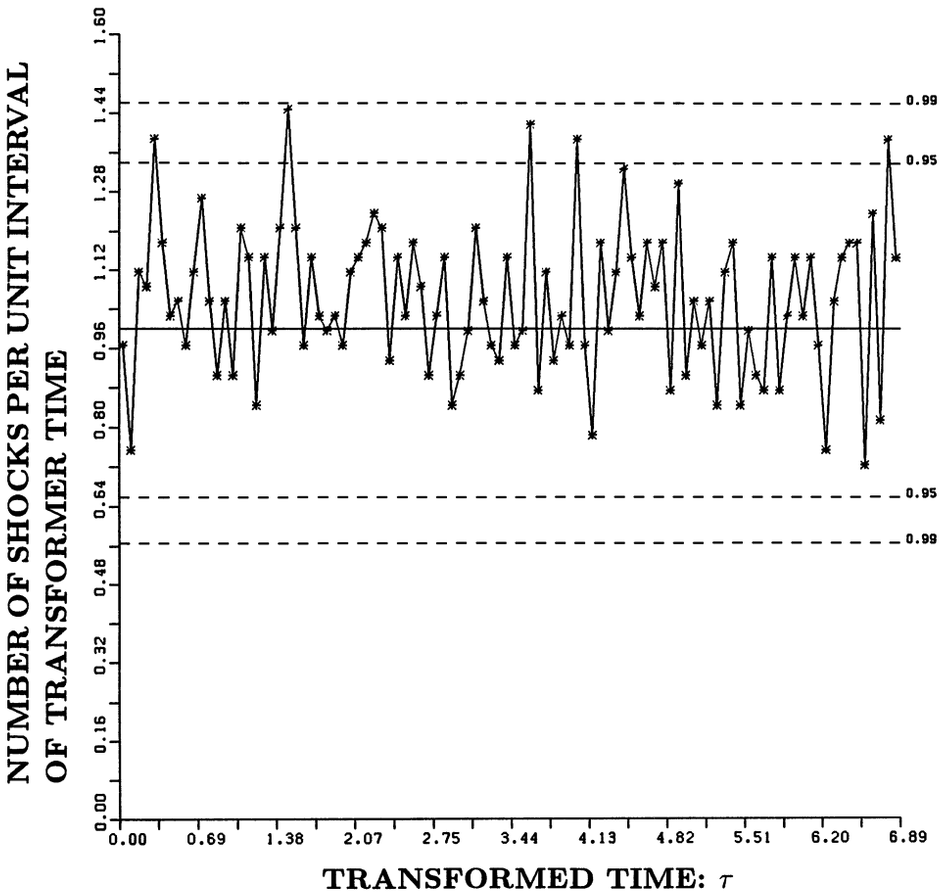



*Figure 14. Histogram for Estimating $M_t(\tau)$ of the Residual Process. The dotted lines indicate 95.5% and 99.7% error bounds for individual estimates assuming the stationary Poisson process.*

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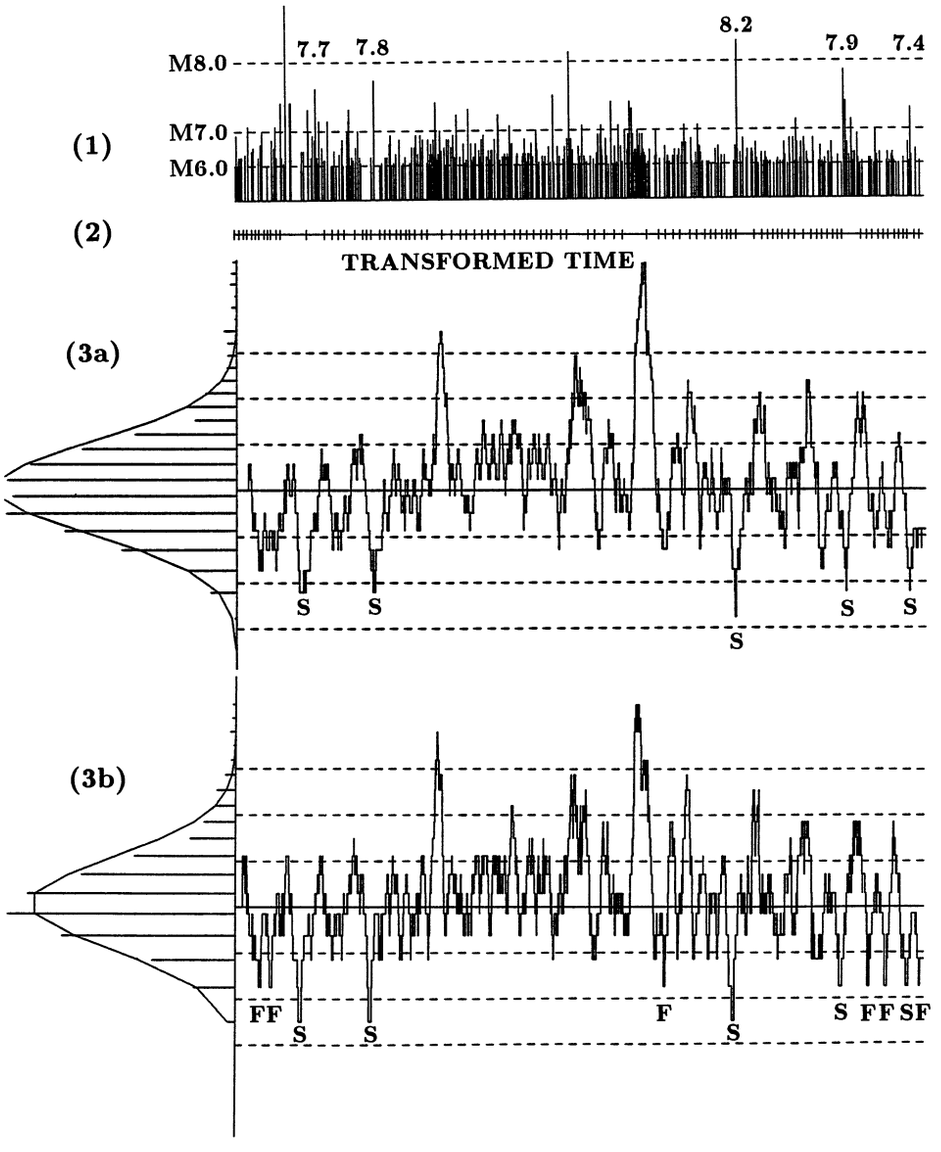



*Figure 15. (1) The magnitudes versus occurrence times of the residual process. (2) The transformed times of January 1 of every year from 1885 to 1980. (3a) Time series of the number of points $\Delta N_\tau$ in the interval $(\tau\text{-}h, \tau)$ of the residual process. Putting $h = 8$, $\Delta N_\tau$ is transformed into $\xi_\tau$ by the relation (27). The dotted lines are one-, two-, and three-fold standard errors of the approximated normal distribution. The sideways histograms indicate the total lengths of the time span of $\tau$ with the same values of $\Delta N_\tau$, and the broken line corresponds to the theoretical distribution (approximate normal) assuming that the residual process is the stationary Poisson process. (3b) Time series of $\xi_\tau$ for $h = 5$.*

of frequency of shocks) of the remaining part seems to be well described by the selected model, with estimated parameters as given in Table 3.

## 4. SEISMIC QUIESCENCE

In contrast to the assertion of Lomnitz and Nava (1983) that seismic quiescence is merely a consequence of the decaying activity of aftershocks, there are some reports discussing the presence of quiescence with quantitative observation of earthquake frequencies. Okada (1978) found some seismicity lowering prior to major earthquakes; that is, the cumulative curve of felt earthquake frequency per month seems to bend lower than the frequency predicted by the combination of the constant rate background seismicity and the series of aftershock-type activity. I am interested here in solving the point at issue by using residual analysis. First of all, from Figure 12 we cannot find a statistically exceptionally long interval $Y_i = \tau_i - \tau_{i-1}$ under the assumption of a stationary Poisson process with intensity rate 1. In addition, we cannot find significant lowering of the number $\Delta N = N(\tau - 8, \tau)$ in Figure 15(3a); there are five times when the trajectory of $\xi = \xi(\Delta N, 8)$ defined in (27) crosses the second-lowest error bound (i.e., $\xi = -2$) from above, but this does not seem unusual for the stationary Poisson process with sample size 483. Do these observations support the aforementioned assertion of Lomnitz and Nava (1983)?

If we look at Figure 15(3a) in comparison with the plot of the transformed data $\{(\tau_i, M_i)\}$ we can see that the five large shocks with $M \geq 7.4$ (i.e., 7.7, 7.8, 8.2, 7.9, and 7.4) took place within a year after $\xi_\tau$ had crossed the level $\xi = -2$ from below. It seems unlikely that such coinci-

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dence is wholly due to the choice of $h$ and $\xi$. For example, a similar plot was made for $h = 5$ in Figure 15(3b), where we see that the three large shocks with $M \geq 7.7$ (i.e., 7.7, 7.8, and 8.2) occurred within a year after $\xi_\tau$ had crossed the level $\xi = -2$ from below and that five shocks with $M \geq 7.4$ (the same as before) occurred within a year after the level crossing of $\xi = -1.5$, which took place 11 times. Are these combinatorial realizations rare phenomena or not? This relates to the question of usefulness of seismic quiescence for predicting the major earthquakes. I tentatively *define* the *quiescence* as the time span of $\tau$ or $t$ when the trajectory of $\xi_\tau$ is lower than a certain level, such as $-2.0$ or $-1.5$, like the above.

Before assessing the probability of the aforementioned events, we have to make assumptions about the stochastic law governing the magnitudes of shocks. From Figure 3 we may assume that the marginal distribution of magnitudes throughout the 98 years in the area is negative exponential. But what should we assume about the joint distribution of magnitudes and occurrence times? Recall here the independence assumptions used to obtain the full likelihood in (22). As mentioned, in studying aftershock sequences, Lomnitz (1966) put forward a hypothesis of "magnitude stability," that is to say, stationarity of the magnitudes and their independence from the occurrence times; it was shown in his paper that the local averages of *aftershock* magnitudes fluctuated slightly about a mean value. This hypothesis was implicitly supported by Utsu (1962) in his detailed trend analysis of three Alaskan aftershock sequences.

Now can this hypothesis extend to the general sequences of the mixture of *main shock* and *aftershocks* in Figure 2? This question relates to the explanation of so-called Båth's law. Båth's law claims the different mean magnitudes between the group of main shocks and group of aftershocks, and Lomnitz and Nava (1983) interpreted this by the different $b$-values [see Eq. (3)] between the two groups. On the other hand, Vere-Jones (1966, 1975) carefully checked such differences using Utsu's table (Utsu 1961) and concluded that Båth's law can be adequately explained by the simple assumption that the magnitudes in an earthquake sequence form a random sample independently selected from a distribution having the exponential form. In addition, the experiment by Utsu (1971, pp. 425–426) suggests that the Gutenberg–Richter law of magnitude frequency with the same $b$-value throughout the whole seismic sequence can imply significantly different $b$-values between the group of main shocks and the group of aftershocks.

It was concluded in Section 3 that the residual time process is a stationary Poisson process. Now a joint model for both magnitude and time will be discussed. This discussion is based on the assumption that one $b$-value applies to the full shock sequence. If we assume that independence of the magnitudes $M_i$ from the past occurrence times of shocks and their magnitudes, $\{(t_j, M_j); t_j < t_i\}$ as well as stationarity, then the residual process $\{(\tau_i, M_i)\}$ is approximately a stationary *compound Poisson process*. Thus we can now assess the probability of the realized combination of the seismic quiescences and the succeeding occurrence of a major earthquake within a year. Since the number of shocks with $M \geq 7.7$ is six in 96 years, 6/96 (per year) may be given for the rate of such shocks. Similarly, 19/96 and 47/96 per year are the rates of shocks with $M \geq 7.4$ and $M \geq 7.0$, respectively. I observed the numbers of the previously defined quiescences in Figure 15 and also counted the major shocks of the specified magnitudes that occurred within a year after the end of the quiescences. Using all of these observations the probabilities of the realized combinations were assessed. These are listed in the row for (i) of Table 4. It is seen that all of the assessed probabilities are very small.

On the other hand, I generated the five sequences (ii)–(vi) of the point process [see Fig. 16 and Ogata (1981) for the simulation method], using the estimated model (15) with the same magnitudes as the original data given in chronological order, and then calculated the probabilities of the realizations of the same combinations as before for each realized residual process. Rows (ii)–(vi) of Table 4 show the observed combinations and the assessments of their probabilities, which exhibit the ordinary probabilities as expected. The implication of the observed small probability events is that the magnitude distribution is *dependent* on the past occurrence times of shocks. In other words, seismic quiescence as defined previously can provide useful information for predicting a major earthquake.

Finally, the stability of the residual process was examined from the predictive viewpoint. The data were divided into two parts, and the earlier part (1885–1949) was used to fit the selected model. Estimated values of parameters, for comparison with Table 3, were $\mu = .00662$, $K = .01769$, $c = .01928$, $p = 1.0$, and $\beta = 1.51912$ with the

*Table 4. Probability Assessments of Realized Combinations*

| Case outcome | $M \geq 7.4$ ($\xi < -2$), S:F | $M \geq 7.7$ ($\xi < -2$), S:F | $M \geq 7.4$ ($\xi < -1.5$), S:F |
|---|---|---|---|
| These data (i) | 5:0 (.00030) | 3:0 (.00024) | 5:6 (.0060) |
| Simulated (ii) | 0:2 (.64) | — | 2:8 (.34) |
| Simulated (iii) | 2:2 (.18) | 0:1 (.94) | 4:13 (.25) |
| Simulated (iv) | 0:3 (.52) | — | 3:11 (.31) |
| Simulated (v) | 2:3 (.26) | 1:4 (.28) | 3:13 (.42) |
| Simulated (vi) | 1:6 (.79) | 0:2 (.88) | 1:16 (.87) |

NOTE: *S* indicates the number of cases in which major earthquakes with magnitudes of the given inequalities took place within a year after the end of the defined quiescences in the text. *F* indicates the number of cases in which the major shock did not take place. The assessed probabilities for the outcomes (*S:F*) are shown in parentheses.

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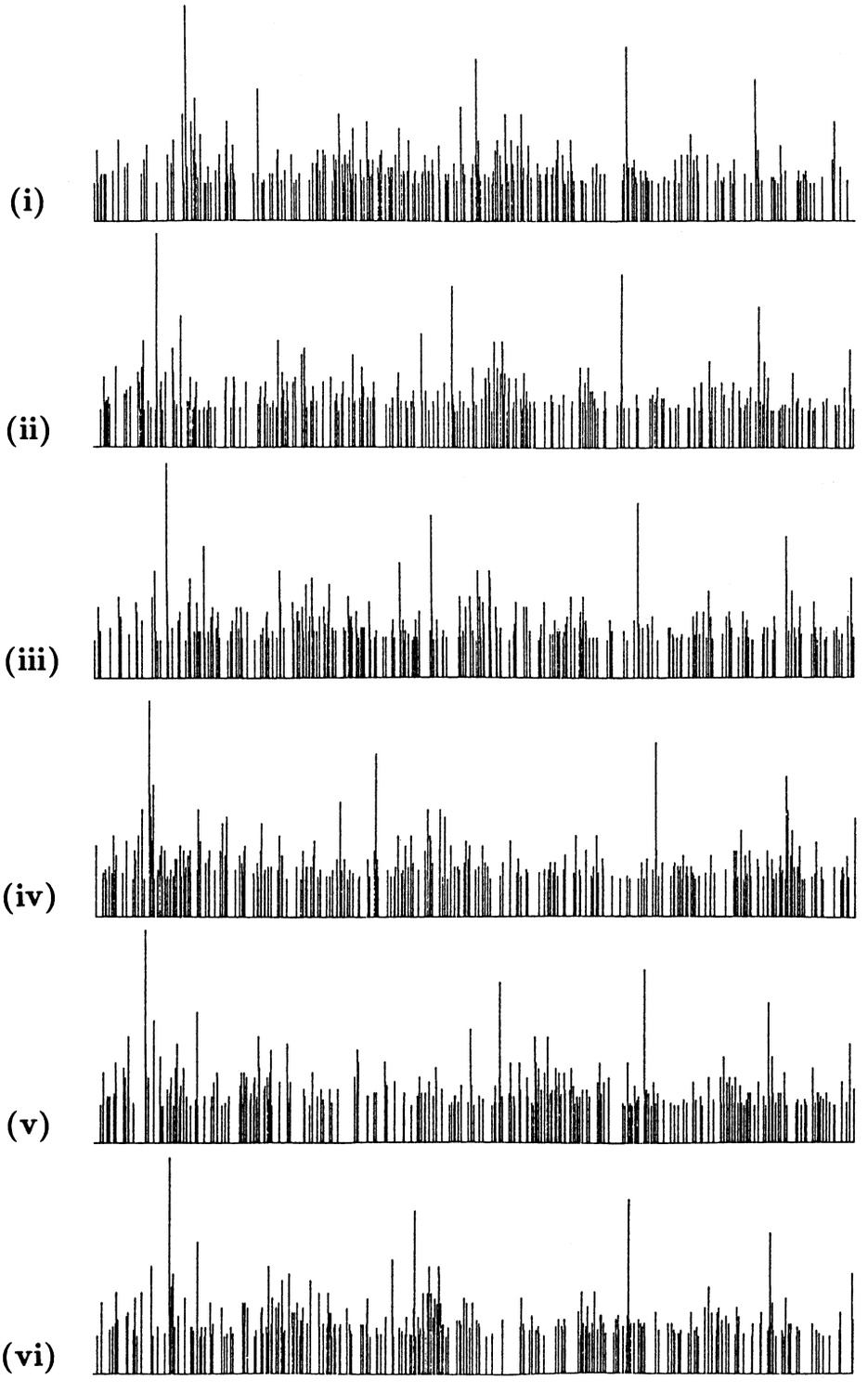



*Figure 16. The Magnitudes Versus Occurrence Times Plot. (i) Real data; (ii)–(vi) simulated data. The order of the magnitudes are the same throughout (i)–(vi).*

reference magnitude $M_r = 6.0$. Then Figure 17, corresponding to Figure 15(3a), is obtained for 1885–1980, in which similar features are seen. In 1951, especially, remarkable quiescence is observed before the large shock ($M8.2$) in 1952. In the period 1950–1980 there are four cases in which $\xi = \xi(\Delta N, 8)$ crosses the level $\xi = -2$ from below, within one year after which three large shocks with $M > 7.4$ occurred.

## 5. CONCLUSION AND SOME REMARKS

We have seen that the epidemic-type model that includes the effect of magnitude gave a better fit to the data than any of the restricted trigger models. This model is defined in terms of the conditional intensity rate, or seismic risk function of time, based on the following simple assumptions: (a) The background seismic activity is gen-

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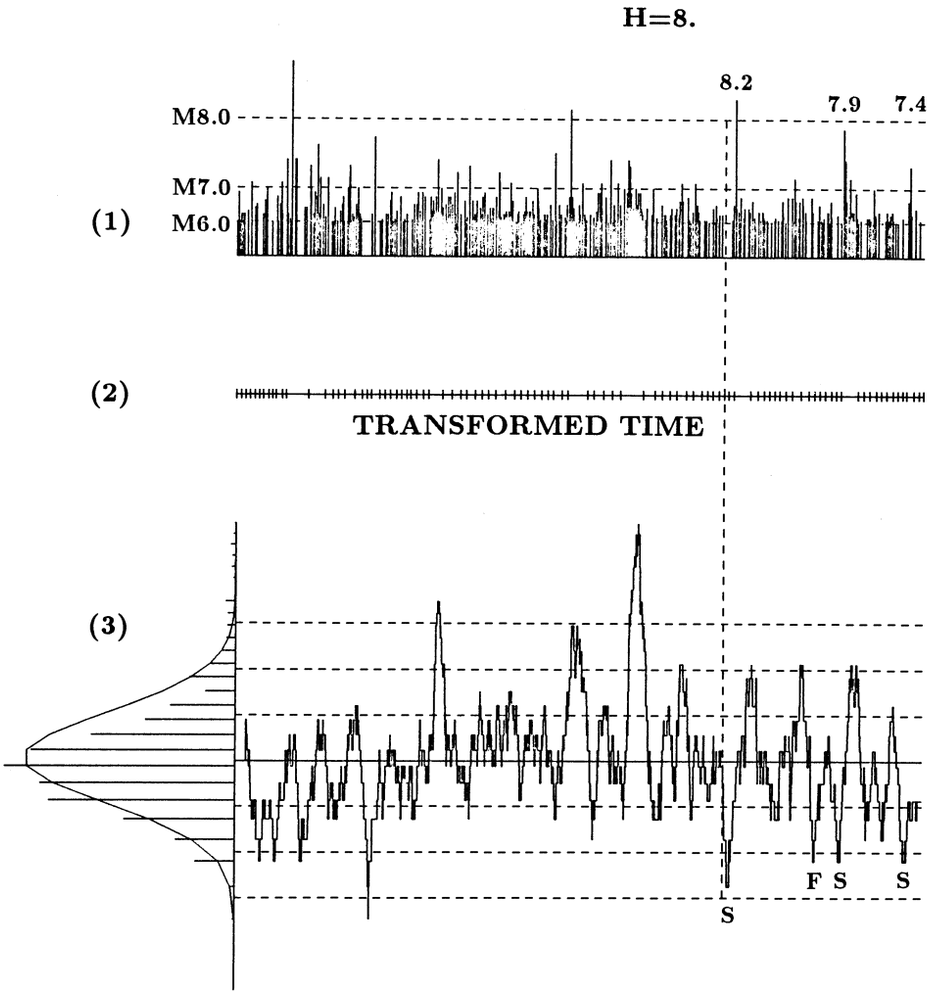



*Figure 17. (1) Magnitudes Versus Occurrence Times of the Residual Process Obtained by the Selected Model for the First Part of the Data (1885–1949). The vertical dotted line shows the end of 1949. (2) and (3) are similar to (2) and (3a) in Figure 15 but are extrapolated for the last part (1950–1980).*

erated by a stationary Poisson process with a constant hazard rate. (b) Each shock has a risk of stimulating aftershocks proportional to $e^{\beta M}$, where $M$ is the magnitude of the shock. (c) The hazard rate of aftershocks decreases with time according to the modified Omori law, $K/(t + c)^p$.

A new method of residual analysis for a modeled point process was developed. This is based on the change of time scale (25) using the estimated conditional intensity (seismic risk function of time). I investigated whether the transformed time-scale version of the data is distributed as a compound Poisson process. The residual analysis for seismicity in the Off Tohoku area indicated that the Poisson model in the changed time scale seems to be acceptable, except during a time period with untypical characteristics. The magnitude distribution, however, is *not* independent of the history of occurrences, especially just after seismic quiescences. In other words, seismic quiescence can be useful for predicting a coming major earthquake.

There were two major earthquakes of the $M8$ class, that is, Off Sanriku 1896 ($M8.5$) and 1933 ($M8.1$), prior to which I did not find quiescence in Figure 15. Both of these earthquakes have unusual characteristics compared with the other major shocks. The 1896 earthquake was a "tsunami earthquake" (Kanamori 1972), which is characterized by an unusually large amount of low-frequency waves. Because of this, the estimated instrumental magnitude is only 6.8 (Utsu 1979), although Abe (1979) gave a magnitude 8.6 estimated from the size of the tsunami. The 1933 Sanriku ($M8.1$) earthquake has been characterized as having the "normal-fault" type of mechanism; that is to say, it was probably caused by gravitational pull exerted by the sinking lithosphere (Kanamori 1971), whereas most of the other major shocks in the east off the Tohoku area are caused by compressive forces. Excepting these, we are left with the striking conclusion that the mechanism underlying very large shocks can affect beforehand the seismicity of a broad area.

I also made a plot of $\xi$, similar to that in Figure 15 for only the main shocks identified by Utsu (see the data in Table 1). If we assume the stationary Poisson model for these shocks in the original time scale, we obtain results similar to those of the previous section. It is generally a

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very hard task, however, to carry out such identification objectively by using the available earthquake catalog (see the last two paragraphs in Sec. 2.1).

Finally, many seismologists believe that there is a *spatial* feature of the earthquake process, known as the *seismic gap*, which is related to the seismic quiescence. This feature, if identified early, could be used not only for the prediction of the location of a large shock in space, but in magnitude as well as in time: A large gap can be expected to be associated with a strong earthquake (see, e.g., Ohtake, Matumoto, and Latham 1977). In the present data set, I have so far not succeeded in identifying seismic gaps. This might be due to the relatively small number of shocks compared with the size of the area. Even if we have a more extensive data set, I believe that such identification would be difficult without suitable modeling of the space–time properties. For example, Motoya (1984) investigated the seismic activity of the whole Hokkaido area and its vicinity, using the catalog of micro-earthquakes (1977–1984) compiled by Hokkaido University. He divided the area into 42 regions and constructed a monthly time series of the percentage of the numbers of regions whose micro-earthquake activity was lower that month than their historical rates. In spite of his success in finding four clear quiescences in the time series just before large shocks, he could not identify the seismic gaps. The Hokkaido area is very active seismically, so the data set is complex. In addition, Motoya's methods are less sophisticated than mine. Using data for all of Japan and analyzing aftershocks instead of main shocks with the techniques of this article, Matsu'ura (1986) recently found 18 quiescences in 11 aftershock sequences before unusually large aftershocks not smaller by more than 1.2 units of magnitude than the main shock. Some space–time plots of hypocenters in her paper show that the recovered activity after the quiescence tends to cluster near the hypocenter of the forthcoming large aftershock. Of course, determining the seismic gap for a large aftershock is easier than for a main shock, because knowledge of the hypocenter of the main shock identifies the general area in which large aftershock might occur. In addition, we may need some additional prior information, such as stress release or accumulation in each focal region, to identify the gaps for the main shocks over a wide area.

Another task for the future is the modeling of the mutual relationship between the occurrence times of the earthquake series and the corresponding magnitudes. It was indicated that the full likelihood for the relation is not as written in (22). The proper model involves a kind of feedback system that is not easily modeled.

Some Fortran programs for the analysis of point processes, including graphical residual analysis, are available in the program package TIMSAC-84 (Akaike et al. 1984, part 2). The other programs used in this article are now being prepared for publication by Ogata and Katsura.

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